

Cylicins are a structural component of the sperm calyx being indispensable for male fertility in mice and human

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Abstract

Cylicins are testis-specific proteins, which are exclusively expressed during spermiogenesis. In mice and humans, two Cylicins, the gonosomal X-linked Cylicin 1 (*Cylc1/CYLC1*) and the autosomal Cylicin 2 (*Cylc2/CYLC2*) genes have been identified. Cylicins are cytoskeletal proteins with an overall positive charge due to lysine-rich repeats. While Cylicins have been localized in the acrosomal region of round spermatids, they resemble a major component of the calyx within the perinuclear theca at the posterior part of mature sperm nuclei. However, the role of Cylicins during spermiogenesis has not yet been investigated. Here, we applied CRISPR/Cas9-mediated gene-editing in zygotes to establish *Cylc1*- and *Cylc2*-deficient mouse lines as a model to study the function of these proteins. *Cylc1* deficiency resulted in male subfertility, whereas *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-}, and *Cylc1*^{-/-} *Cylc2*^{-/-} males were infertile. Phenotypical characterization revealed that loss of Cylicins prevents proper calyx assembly during spermiogenesis. This results in decreased epididymal sperm counts, impaired shedding of excess cytoplasm, and severe structural malformations, ultimately resulting in impaired sperm motility. Furthermore, exome sequencing identified an infertile man with a hemizygous variant in *CYLC1* and a heterozygous variant in *CYLC2*, displaying morphological abnormalities of the sperm including the absence of the acrosome. Thus, our study highlights the relevance and importance of Cylicins for spermiogenic remodeling and male fertility in human and mouse, and provides the basis for further studies on unraveling the complex molecular interactions between perinuclear theca proteins required during spermiogenesis.

eLife assessment

This study provides **valuable** insights into the role of two under-researched sperm-specific proteins (Cylicin 1 and Cylicin 2). The authors provide **convincing** evidence that they have an essential role in sperm head structure during spermatogenesis, and that their loss leads to subfertility or infertility, with a dose-dependent phenotype. Importantly, the authors identify infertile males with mutations in both Cylicin1 and Cylicin2. Thus, the findings from the mouse models might be applicable to understanding human male infertility with similar structural defects.

Introduction

The differentiation of round spermatids into sperm during spermiogenesis is a highly organized and spatiotemporally controlled process taking place in the seminiferous epithelium of the testis. Cellular and morphological remodeling involves DNA hypercondensation, establishment of a species-specific head morphology, removal of excess cytoplasm as well as formation of accessory structures like acrosome and flagellum. These structural changes depend on a highly efficient protein trafficking machinery and a unique sperm cytoskeleton¹. One essential cytoskeletal element is the perinuclear theca (PT), which surrounds the sperm nucleus, except for the caudal edge, at the implantation site of the flagellum. The PT is supposed to serve as a structural scaffold for the sperm nucleus and resembles a rigid cytosolic protein layer, which is resistant to non-ionic detergents and high salt buffer extractions^{2,3}. The PT has been subdivided into a subacrosomal and postacrosomal part based on its localization, function, composition, and developmental origin. The subacrosomal part of the PT develops early during spermiogenesis, simultaneously with the formation of the acrosome. It is supposed to emerge from acrosomal vesicles, and presents as a thin cytosolic protein layer between the inner acrosomal membrane and the nuclear envelope^{4,5}. The postacrosomal part of the PT, also known as the sperm calyx, originates from cytosolic proteins that are transported via the manchette and assembled during sperm head elongation⁴. It is located in between the nuclear envelope and the sperm plasma membrane.

Proteomic analyses using murine and bovine PT extracts revealed 500-800 different proteins, highlighting its high molecular complexity^{6,7}. Apart from cytoskeletal proteins, signaling molecules and several *de novo* synthesized core histones were identified. They are proposed to be essential for sperm-egg interaction and chromatin remodeling of the male pronucleus at early post-fertilization stages^{8,9,4,10,11}. Many structural proteins of the PT are testis-specific and uniquely expressed in the PT, including Calicin (*Ccin*)^{12,2,13}, Cylicin 1 (*Cylc1*) and Cylicin 2 (*Cylc2*)^{2,14,15}, actin-capping proteins CPβ3 and CPα3¹⁶⁻¹⁸ as well as actin-related proteins Arp-T1 and Arp-T2¹⁹. Despite being characterized on molecular level, their function remains largely elusive.

Both, *Ccin* and Cylicins are highly basic proteins with a predominant localization in the calyx of mature sperm^{14,15,12}. In most species, two Cylicin genes, *Cylc1* and *Cylc2*, have been identified (Figure 1-supplement 1). They are characterized by repetitive lysine-lysine-aspartic acid (KKD) and lysine-lysine-glutamic acid (KKE) peptide motifs, resulting in an isoelectric point (IEP) > pH 10^{14,15}. Repeating units of up to 41 amino acids in the central part of the molecules stand out by a predicted tendency to form individual short α-helices¹⁴. Mammalian Cylicins exhibit similar protein and domain characteristics, but CYLC2 has a much shorter amino-terminal portion than CYLC1 (Figure 1 - supplement 1). While the *CYLC2/Cylc2* gene is encoded on autosomes, the human and murine *CYLC1/Cylc1* gene is encoded on the X-chromosome, resulting in hemizygoty

in males. Further, in bovine, *Cylc2* serves as actin-binding protein of the sperm perinuclear cytoskeleton²⁰. Cylicins seem to be cytoskeletal regulators and required for proper sperm head architecture.

In this study, we report the CRISPR/Cas9-mediated generation and characterization of *Cylc1*-, *Cylc2*-, and *Cylc1/2*-deficient mice, demonstrating that Cylicins are indispensable for male fertility and play a key function in the formation of the sperm calyx. We show that Cylicins are required for the maintenance of the PT integrity during spermiogenesis and in mature sperm. Their deficiency results in morphological defects of the sperm head, acrosome, and mid-piece. Our analyses revealed that *Cylicin* genes are evolutionary conserved in rodents and primates. Furthermore, we identified Cylicin variants in an infertile man, demonstrating the conserved role of Cylicins in regulating male infertility.

Results

Cylicins are indispensable for male fertility in mice

Cylicins were first discovered and characterized in the early nineties due to their subcellular localization a role in sperm head architecture was postulated^{2,14,15}. To address the question of the role of Cylicins during spermiogenesis, we used CRISPR/Cas9-mediated gene-editing to generate *Cylc1*- and *Cylc2*-deficient mouse models. First, for *Cylc1*, a pair of sgRNAs targeting exon 5 was designed, and a mouse line with a frameshift inducing deletion of 1.585 kb, accounting for 85% of the *Cylc1* coding sequence, was established (**Fig. 1 A**). Next, two guide RNAs targeting exon 4/5 of the *Cylc2* gene were applied to establish a *Cylc2*-deficient mouse line with a 1.145 kb frameshift inducing deletion (**Fig. 1 A**). In both lines, the majority of predicted functional domains with repetitive, lysine-lysine-aspartic acid (KKD) and lysine-lysine-glutamic acid (KKE) peptide motifs were depleted. Deletion was confirmed by PCR-based genotyping (**Fig. 1 B**).

Fertility testing of *Cylc1*^{-/-} males revealed significantly reduced pregnancy rates (16 %) and mean litter size (2.2) (**Fig. 1 C**). *Cylc2*^{-/-} males were infertile, while *Cylc2*^{+/-} males showed no significant difference in fertility parameters compared to wildtype mice (**Fig. 1 C**). Additionally, established mouse lines were intercrossed to generate *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} males. Of note, *Cylc1*^{-/-} *Cylc2*^{+/-}, and *Cylc1*^{-/-} *Cylc2*^{-/-} males were infertile (**Fig. 1 C**). This indicates that loss of *Cylc1* alone is partially tolerated, as suggested by subfertility of *Cylc1*^{-/-} males, whereas the additional loss of one *Cylc2* allele renders male mice infertile. Taken together, the results suggest, that two functional Cylicin alleles are required for male fertility.

qRT PCR confirmed the absence of *Cylc1* and/or *Cylc2* transcripts in Cylicin-deficient animals (**Fig. 1 D**). In *Cylc2*^{+/-} animals expression of *Cylc2* was reduced by 50 %. Neither loss of *Cylc1* nor *Cylc2* resulted in upregulation of *Cylc2* or *Cylc1*, respectively.

Next, due to the lack of commercial antisera, polyclonal antibodies against murine CYLC1 and CYLC2 were raised to visualize the localization of Cylicins during spermiogenesis. Specificity of antibodies was proven by immunohistochemical stainings (IHC), showing a specific signal in testis sections only, but not in any other organ tested (Figure 1 - supplement 2). Immunofluorescence staining of wildtype testicular tissue showed presence of both, CYLC1 and CYLC2 from the round spermatid stage onward (**Fig. 1 E**). The signal was first detectable in the subacrosomal region as a cap-like structure, lining the developing acrosome (**Fig. 1 E-F**, Figure 1 – supplement 3). As the spermatids elongate, CYLC1 and CYLC2 move across the PT towards the caudal part of the cell (Figure 1 – supplement 4). At later steps of spermiogenesis, the localization in the subacrosomal part of the PT faded, while it intensified in the postacrosomal calyx region (**Fig. 1 E-F**). Of note, the localization of CYLC1 and CYLC2 in the calyx of mature sperm has been reported in bovine and

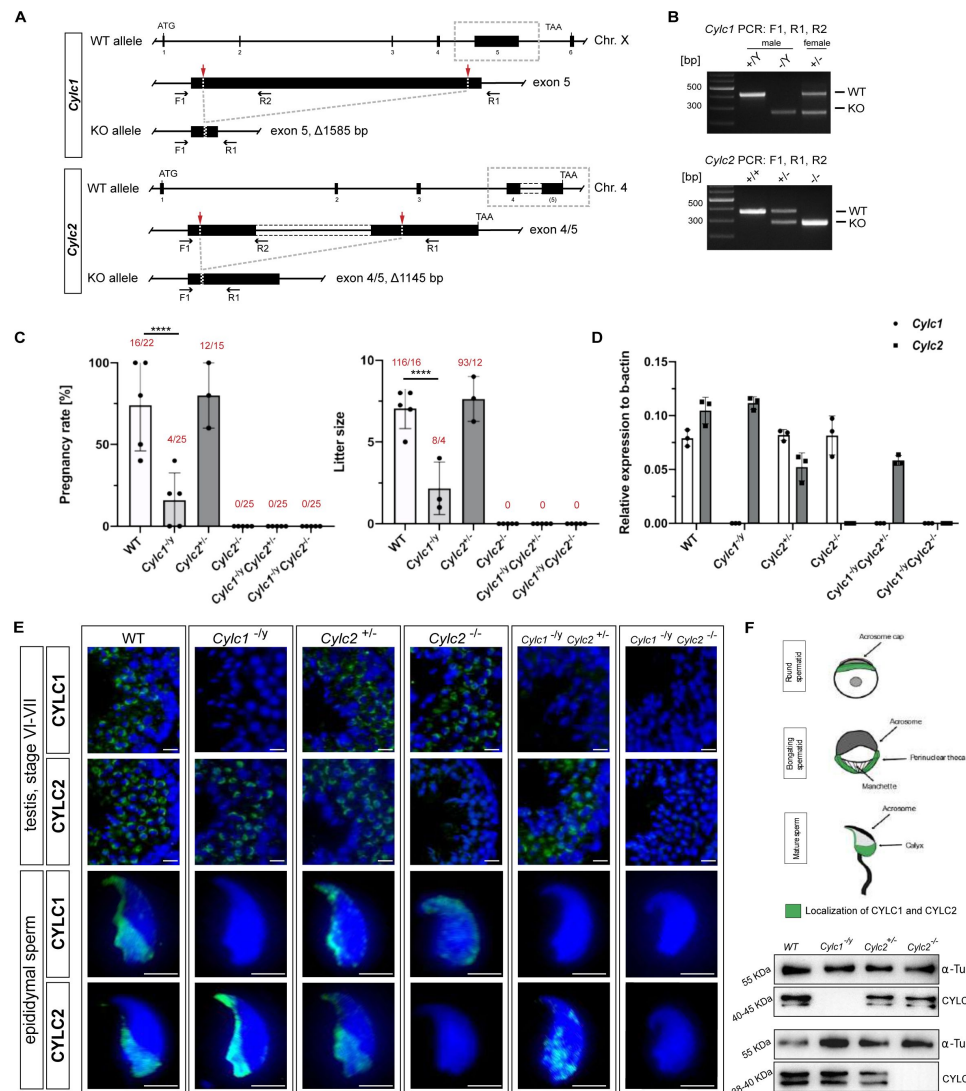


Figure 1

Loss of *Cylc1* or *Cylc2* results in impaired male fertility.

(A) Schematic representation of the *Cylc1* and *Cylc2* gene structure and targeting strategy for CRISPR/Cas9-mediated generation of *Cylc1*- and *Cylc2*-deficient alleles. Targeting sites of guide RNAs are depicted by red arrows. Genotyping primer binding sites are depicted by black arrows.

(B) Representative genotyping PCR of *Cylc1*- and *Cylc2*-deficient mice. N=3

(C) Fertility analysis of *Cylc1*-deficient mice visualized by mean litter size and pregnancy rate (%) in comparison to wildtype matings. Black dots represent mean values obtained for each male included in fertility testing. Columns represent mean values $\pm$ standard deviation (SD). Total number of offspring per total number of pregnancies as well as total number of pregnancies per total number of plugs are depicted above each bar.

(D) Expression of *Cylc1* and *Cylc2* in testicular tissue of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} mice analyzed by qRT-PCR. Biological replicate of 3 was used.

(E) Immunofluorescent staining of testicular tissue and cauda epididymal sperm from WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} males against CYLC1 and CYLC2. Cell nuclei were counterstained with DAPI. Staining was performed on 3 animals from each genotype. Scale bar: 5 μ m.

(F) Schematic illustration of CYLC localization during spermiogenesis. CYLC localization (green) is shown for round and elongating spermatids as well as mature sperm.

(G) Representative immunoblot against CYLC1 and CYLC2 on cytoskeletal protein fractions from WT, *Cylc1*^{-/-}, *Cylc2*^{+/-} and *Cylc2*^{-/-} testes. α -Tubulin was used as load control.

human. The generated antibodies did not stain testicular tissue and mature sperm of *Cylc1*- and *Cylc2*-deficient males, except for a very weak unspecific background staining in the lumen of seminiferous tubules and the residual bodies of testicular sperm (**Fig. 1 E**). Additionally, Western Blot analyses confirmed the absence of CYLC1 and CYLC2 in cytoskeletal protein fractions of the respective knockout (**Fig. 1 G**), thereby demonstrating i) specificity of the antibodies and ii) validating the gene knockout. Of note, the CYLC1 antibody detects a double band at 40-45 KDa. This is smaller than the predicted size of 74 KDa as, but both bands were absent in *Cylc1*^{-/-}. Similarly, the CYLC2 Antibody detected a double band at 38-40 KDa instead of 66 KDa. Again, both bands were absent in *Cylc2*^{-/-}. Next, Mass spectrometry analysis of cytoskeletal protein fraction of mature spermatozoa was performed detecting both proteins in WT but not in the respective knockout samples (Figure 1 – supplement 5; Figure 1 – supplement 6).

Sperm morphology is severely altered in Cylicin-deficient mice

Next, spermiogenesis of Cylicin-deficient males was analyzed in detail. Gross testicular morphology as well as testicular weight were not significantly altered (**Fig. 2 A, B**). The testicular morphology appeared unaltered, with all stages of spermatogenesis being detectable in hematoxylin and eosin (HE) stained testicular sections (Figure 2 – supplement 1). However, a strong decline in cauda epididymal sperm counts was observed in all Cylicin-deficient males. For *Cylc1*^{-/-} and *Cylc2*^{+/-} males, a moderate reduction of 40-47% was determined, whereas *Cylc2*^{-/-} and *Cylc1*^{-/-} *Cylc2*^{+/-} displayed an approx. 65% reduction in epididymal sperm counts compared to wildtype mice (**Fig. 2 C**). In *Cylc1*^{-/-} *Cylc2*^{-/-} males, spermiogenesis was most impaired, as indicated by an 85% reduction of the sperm count (**Fig. 2 C**). Eosin-Nigrosin staining revealed that the viability of epididymal sperm from all genotypes was not severely affected (**Fig. 2 D**, Figure 2 – supplement 2). However, viability of *Cylc2*^{-/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} sperm was significantly reduced by approx. 15% compared to wildtype sperm. (**Fig. 2 D**).

Next, we used brightfield microscopy to evaluate the effects of Cylicin deficiency on sperm morphology. Above all, coiling of the sperm tails and kinked sperm heads were observed in *Cylc2*^{-/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} males (**Fig. 2 E**). To confirm this, we used PNA-FITC lectin immunofluorescence staining to analyze acrosome localization in mature sperm, MITOred to visualize mitochondria in the flagellum and DAPI to stain the nucleus (**Fig. 2 F**). Loss of *Cylc1* alone caused malformations of the acrosome in around 38% of mature sperm, while their flagellum appeared unaltered and properly connected to the head. *Cylc2*^{+/-} males showed normal sperm tail morphology with around 30% of acrosome malformations. *Cylc2*^{-/-} mature sperm cells displayed morphological alterations of head and mid-piece (**Fig. 2 F-G**). 76% of *Cylc2*^{-/-} sperm cells showed acrosome malformations, bending of the neck region, and/or coiling of the flagellum, occasionally resulting in its wrapping around the sperm head in 80% of sperm (**Fig. 2 F**). While 70% of *Cylc1*^{-/-} *Cylc2*^{+/-} sperm showed these morphological alterations, around 92% of *Cylc1*^{-/-} *Cylc2*^{-/-} sperm presented with coiled tail and abnormal acrosome (**Fig. 2 F-G**).

To analyze in detail the sperm head, we used Nuclear Morphology software on DAPI stained sperm samples. *Cylc1*^{-/-} and *Cylc2*^{+/-} sperm showed no alterations of the nuclear shape when compared to wildtype (**Fig. 2 H**, Fig. S2 C, D). However, heads of *Cylc2*^{-/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} sperm appeared smaller, with shorter hooks and increased circularity of the nuclei as well as reduced elongation (**Fig. 2 H**, Fig. S2 D). Interestingly, *Cylc1*^{-/-} *Cylc2*^{+/-} sperm heads appeared unaltered, suggesting that only *Cylc2* has a crucial role for sperm head shaping, and one functional *Cylc2* allele is sufficient to maintain the correct shape of the nucleus (**Fig. 2 H**, Figure 2 – supplement 3).

To study the effects of Cylicin deficiency on sperm calyx integrity and morphology, we analyzed the localization of other calyx-specific proteins, such as CCIN and CapZa3. In epididymal sperm, CCIN co-localize with both CYLC1 and CYLC2 in the calyx (Figure 2 – supplement 4). In *Cylc1*^{-/-} and *Cylc2*^{+/-} sperm, CCIN localization remained unchanged, being present in the calyx and in the ventral portion of PT as described previously⁷. However, in 91% of *Cylc2*^{-/-} sperm, CCIN

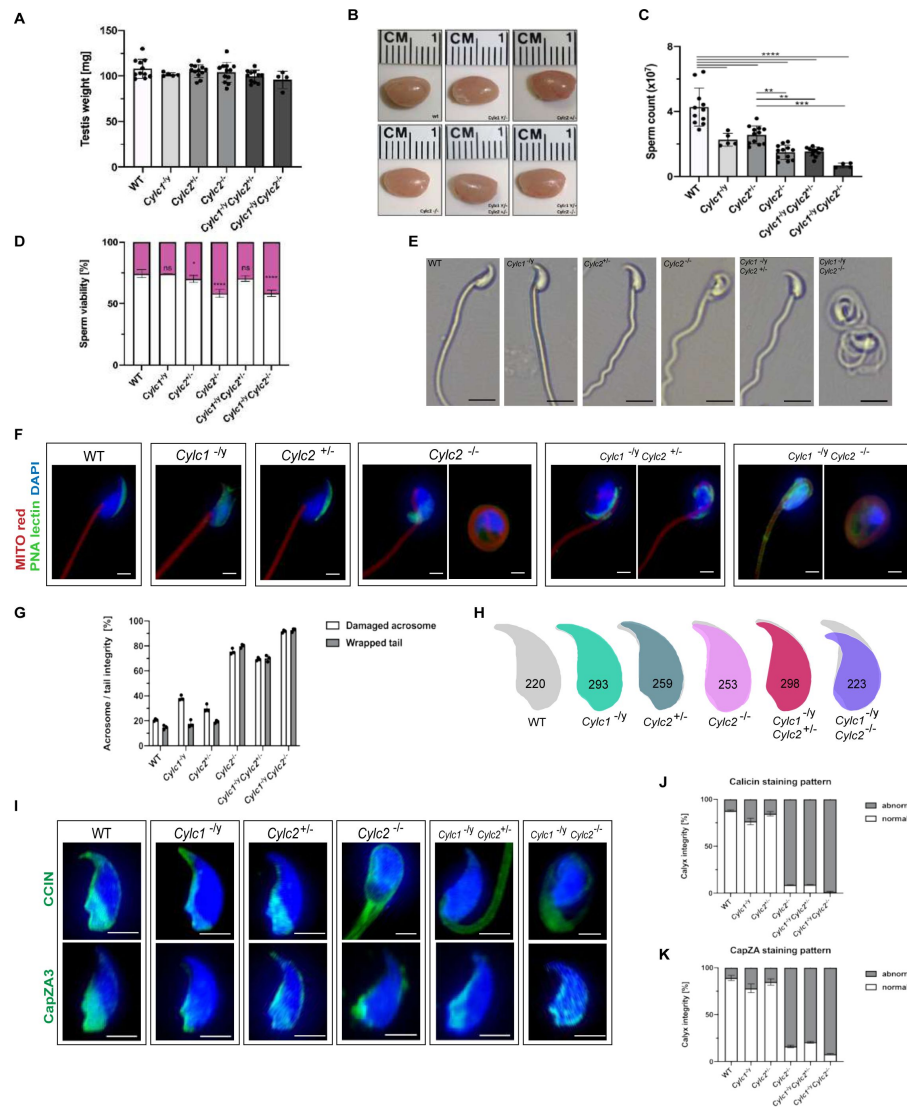


Figure 2

Sperm morphology is severely altered in Cylicin-deficient mice

(A) Testis weight [mg] and sperm count ($\times 10^7$) of WT, *Cylic1*^{-/-}, *Cylic2*^{+/-}, *Cylic2*^{-/-}, *Cylic1*^{-/-} *Cylic2*^{+/-} and *Cylic1*^{-/-} *Cylic2*^{-/-} males. Mean values $\pm$ SD are shown; block dots represent data points for individual males.

(B) Comparable photographs of the testes of WT, *Cylic1*^{-/-}, *Cylic2*^{+/-}, *Cylic2*^{-/-}, *Cylic1*^{-/-} *Cylic2*^{+/-} and *Cylic1*^{-/-} *Cylic2*^{-/-} mice.

(C) Epididymal sperm count ($\times 10^7$) of WT, *Cylic1*^{-/-}, *Cylic2*^{+/-}, *Cylic2*^{-/-}, *Cylic1*^{-/-} *Cylic2*^{+/-} and *Cylic1*^{-/-} *Cylic2*^{-/-} males. Mean values $\pm$ SD are shown; block dots represent data points for individual males.

(D) Viability of the epididymal sperm stained with Eosin Nigrosine. Percentage of Eosin negative (viable) and Eosin positive (inviolate) sperm is shown. Data represented as mean $\pm$ SD. Staining was performed on 3 animals from each genotype.

(E) Bright field microscopy pictures of epididymal sperm from WT, *Cylic1*^{-/-}, *Cylic2*^{+/-} and *Cylic2*^{-/-} mice. Scale bar: 10 μ m.

(F) Immunofluorescence staining of epididymal sperm acrosomes with PNA lectin (green) and tails with MITOred (red). Nuclei were counterstained with DAPI. Scale bar: 5 μ m.

(G) Quantification of abnormal sperm of WT, *Cylic1*^{-/-}, *Cylic2*^{+/-}, *Cylic2*^{-/-}, *Cylic1*^{-/-} *Cylic2*^{+/-} and *Cylic1*^{-/-} *Cylic2*^{-/-} mice is shown. Acrosome aberrations and tail coiling were counted separately. Staining was performed on 3 animals from each genotype.

(H) Nuclear morphology analysis of WT, *Cylic1*^{-/-}, *Cylic2*^{+/-}, *Cylic2*^{-/-}, *Cylic1*^{-/-} *Cylic2*^{+/-} and *Cylic1*^{-/-} *Cylic2*^{-/-} sperm. Number of cells analyzed for each genotype is shown.

(I) Representative pictures of immunofluorescent staining against PT proteins CCIN (upper panel) and CAPZA3 (lower panel) in WT, *Cylic1*^{-/-}, *Cylic2*^{+/-}, *Cylic2*^{-/-}, *Cylic1*^{-/-} *Cylic2*^{+/-} and *Cylic1*^{-/-} *Cylic2*^{-/-} sperm. Nuclei were counterstained with DAPI. Staining was performed on 3 animals from each genotype. Scale bar: 5 μ m.

(J-K) Quantification of sperm with abnormal calyx integrity in WT, *Cylic1*^{-/-}, *Cylic2*^{+/-}, *Cylic2*^{-/-}, *Cylic1*^{-/-} *Cylic2*^{+/-} and *Cylic1*^{-/-} *Cylic2*^{-/-} mice based on CCIN and CapZA staining patterns.

localized to the tail or in random parts of the sperm head (**Fig. 2 I, J**). In 91% of *Cylc1*^{-/-}*Cylc2*^{+/-} and 98% of *Cylc1*^{-/-}*Cylc2*^{-/-} sperm, the localization of CCIN was also significantly altered, with the signal mainly being present in the sperm tail. CapZα3 forms a heterodimer with CapZβ3, creating a functional complex that localizes in the calyx²¹. Immunofluorescence stainings revealed that the localization of CapZα3 remained unchanged in *Cylc1*^{-/-} and *Cylc2*^{+/-} mice compared to wildtype mice. In 84% of *Cylc2*^{-/-} sperm cells, CapZα3 localized in the caudal portion of the head but without regular calyx localization (**Fig. 2 I, K**). Interestingly, *Cylc1*^{-/-}*Cylc2*^{+/-} mice showed less severe anomalies of the calyx and although CCIN was located almost exclusively in the tail, CapZα3 maintained the correct calyx localization in around 30% of sperm (**Fig. 2 I**). Finally, 92% of *Cylc1*^{-/-}*Cylc2*^{-/-} spermatozoa showed CapZα3 localization across the sperm head without regular calyx shape. These results suggest that the loss of Cylicins impairs the formation of calyx and the correct localization of its components, which might contribute to morphological anomalies of the sperm described initially.

***Cylc2*^{-/-} sperm cells have altered flagellar beat**

Transmission electron microscopy (TEM) of epididymal sperm confirmed the severe structural defects observed by light microscopy (**Fig. 3 A**, Figure 3 – supplement 1): *Cylc2*^{-/-} sperm showed coiling of the tail and dislocation of the head-tail connecting piece from the basal plate, resulting in parallel positioning of head and tail (**Fig. 3 A**, white arrowheads). Furthermore, in *Cylc2*^{-/-} sperm, excess of cytoplasm was observed, located around the nucleus and coiled tail (**Fig. 3 A**). Anomalies of the head were observed at the level of the PT, while the nuclei appeared unaltered. In all *Cylc2*^{-/-} sperm cells, the posterior portion of PT – calyx was absent (**Fig. 3 A**, red arrowheads). Instead of surrounding the nucleus entirely, the PT in *Cylc2*^{-/-} sperm appeared interrupted, missing completely its caudal part. Further, we observed loosening of the peri-acrosomal region, which is not compact and adherent to the nucleus (**Fig. 3 A**, green arrowheads). On the contrary, *Cylc1*^{-/-} sperm cells appeared healthy, with intact PT, acrosome and calyx.

While the motility of *Cylc1*^{-/-} and *Cylc2*^{+/-} sperm remained unchanged compared to wildtype sperm (around 60% motile cells), motility of *Cylc2*^{-/-} sperm was drastically reduced to only 7% motile sperm (**Fig. 3 B**) and the motility of *Cylc1*^{-/-}*Cylc2*^{-/-} sperm was reduced to 2% motile sperm. In addition, the few motile sperm cells were not progressive but were swimming in circular trajectories. Interestingly, in *Cylc1*^{-/-}*Cylc2*^{+/-} mice, sperm motility was reduced as well, but less drastically, with 27% of sperm cells being motile (**Fig. 3 B**).

The SpermQ software was used to analyze the flagellar beat of non-capacitated *Cylc2*^{-/-} sperm in detail²². *Cylc2*^{-/-} sperm showed stiffness in the neck and a reduced amplitude of the initial flagellar beat, as well as reduced average curvature of the flagellum during a single beat (Figure 3 – supplement 2). Interestingly, we observed that the flagellar beat of *Cylc2*^{-/-} sperm cells was similar to wildtype cells, however, with interruptions during which midpiece and initial principal piece appeared stiff, whereas high frequency beating occurs at the flagellar tip (**Fig. 3 C**, Video 1, Video 2). These interruptions occurred only on the open-hook side and the duration of such interruptions varied from beat to beat. Of note, similar phenotypes have been observed for sperm with structural defects in the axoneme²³, however axoneme structure of Cylicin-deficient sperm appeared unaltered, presenting typical 9+2 microtubular composition in all genotypes (Figure 3 – supplement 1). Thus, we hypothesize that observed structural defects of the PT and head-tail connecting piece are restrictive for sperm motility and physiological beating patterns.

Taken together, observed anomalies of sperm heads, impaired sperm motility, and the decrease in epididymal sperm concentration show that Cylicin deficiency resembles a severe OAT (Oligo-Asthenozoospermia-syndrome) described in human.

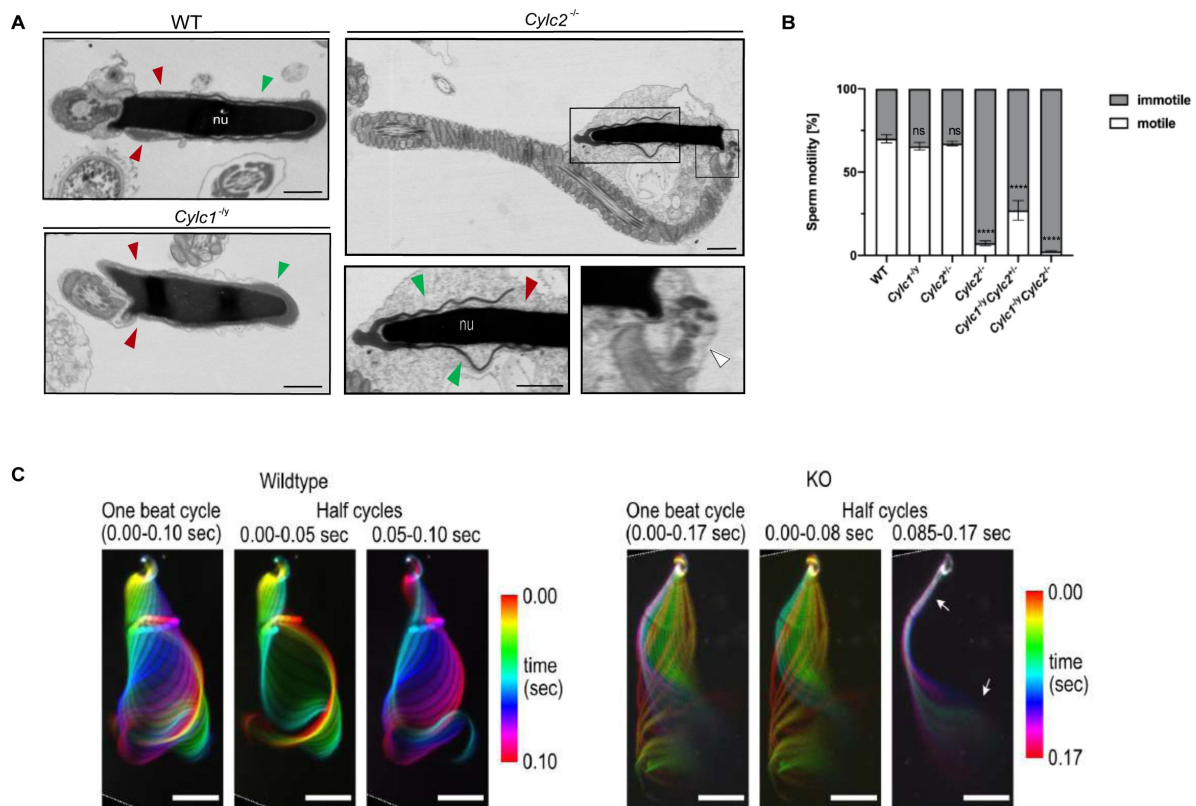


Figure 3

Cylc2^{-/-} sperm cells have altered flagellar beat

(A) TEM micrographs of WT, *Cylc1*^{-/-} and *Cylc2*^{-/-} epididymal sperm. Acrosome appears detached from the nucleus in *Cylc2*^{-/-} sperm (green arrowheads), while the calyx is missing entirely (red arrowheads). The head-tail connecting piece shifted from the basal plate is shown by white arrowheads causing the looping of the flagellum and formation of a cytoplasmic sac. *Cylc1*^{-/-} sperm appears comparable to WT. Scale bar: 1 μ m.

(B) Motility of the epididymal sperm of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} males activated in TYH medium.

(C) Full and half-beat cycle plots of the flagellar beat are shown for WT and *Cylc2*^{-/-} spermatozoa. Half beat cycle shows the stiffness of the midpiece (upper arrow) and high oscillations (lower arrow) in *Cylc2*^{-/-} sperm in one direction of the beat.

Cylicins are required for acrosome attachment to the nuclear envelope

To study the origin of observed structural sperm defects, spermiogenesis of Cylicin-deficient males was analyzed in detail. PNA lectin staining and Periodic Acid Schiff (PAS) staining of testicular tissue sections were performed to investigate acrosome biogenesis. During Golgi phase, the acrosome first starts to appear as an aggregation of proacrosomal vesicles into a single granule. This premature acrosomal structure was unaltered in all genotypes, with PNA signal appearing as a small dot on one pole of round spermatids (**Fig. 4 A-B**). During cap phase, acrosomes grow to cover the apical part of the nucleus. In wildtype and *Cylc2*^{+/-} mice, the forming acrosome appeared equally smooth and showed a regular cap structure on the perinuclear region of round spermatids. However, in some of the round spermatids from *Cylc2*^{-/-} and *Cylc1*^{-/-} mice, gaps in the forming acrosome were observed, as well as an irregular shape of the cap. In *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} mice, most of the round spermatids were deformed or displayed irregularly localized caps (**Fig. 4 A-B**, Figure 4 – supplement 1, Figure 4 – supplement 2). At acrosome phase, many elongating spermatids of *Cylc1*^{-/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} mice had irregular acrosome (**Fig. 4 A-B**, Figure 4 – supplement 1, Figure 4 – supplement 2). Detachment of the acrosome from the nuclear envelope was evident in testis samples of *Cylc2*^{-/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} male mice. These results suggest that Cylicins are required for the attachment of the developing acrosome to the nuclear envelope during spermiogenesis. Microtubules appeared longer on one side of the nucleus than on the other, displacing the acrosome to the side and creating a gap in the PT (**Fig. 4 C**). Whereas elongated spermatids at step 14-15 in wildtype sperm already disassembled their manchette and the PT appeared as a unique structure that compactly surrounds nucleus, in *Cylc2*^{-/-} spermatids, remaining microtubules failed to disassemble (**Fig. 4 C**, yellow arrowhead), and the acrosome detached from the nuclear envelope (green arrowhead). In addition, at step 16, the calyx was absent, and an excess of cytoplasm surrounded the nucleus and flagellum (**Fig. 4 C**, white arrowhead). Furthermore, many damaged and degrading cells were observed in testicular tissue TEM samples, having perforated nuclei and detached structures (Figure 4 – supplement 3). Interestingly, phagosomes with cellular remains were observed far away from the lumen and sometimes even at the basal membrane of the tubuli, suggesting that the cells that suffer most severe structural damage are being removed. This mechanism of removing malformed cells explains the reduction of epididymal sperm count in Cylicin-deficient genotypes.

Cylicin deficiency results in abnormal manchette elongation and disassembly

Next, we investigated the role of Cylicins during formation and development of the manchette – a sperm-specific, transient structure that represents a microtubular platform for protein transport, which showed several anomalies in TEM. Transport of the intracellular vesicles is crucial for the formation of the flagellum, acrosome assembly and removal of excess cytoplasm. The manchette is first detected at step 8 at the perinuclear ring of round spermatids, just prior to their elongation. During the next steps of spermiogenesis, as the spermatids elongate, manchette moves towards the neck region in a skirt-like structure and starts disassembling at step 13 when the elongation is complete²⁴. We used immunofluorescence staining of α -tubulin on squash testis samples containing spermatids at different stages of spermiogenesis to investigate whether the altered head shape, calyx structure, and tail-head connection anomalies originate from defects of the manchette structure. Spermatids starting from step 8 were observed individually for step-to-step comparison. In all genotypes, a cap-like shape of the manchette was observed in step 8 round spermatids, suggesting that the manchette assembles properly and starts elongating towards the neck region during step 9 (Figure 5 – supplement 1). In all genotypes, the typical skirt-like structure was observed at the caudal region of the spermatids at step 10 and 11, suggesting that the manchette assembles correctly even in Cylicin-deficient sperm (**Fig. 5 A**). In spermatids from *Cylc1*^{-/-} and *Cylc2*^{+/-} mice, regular manchette development was observed in further steps of

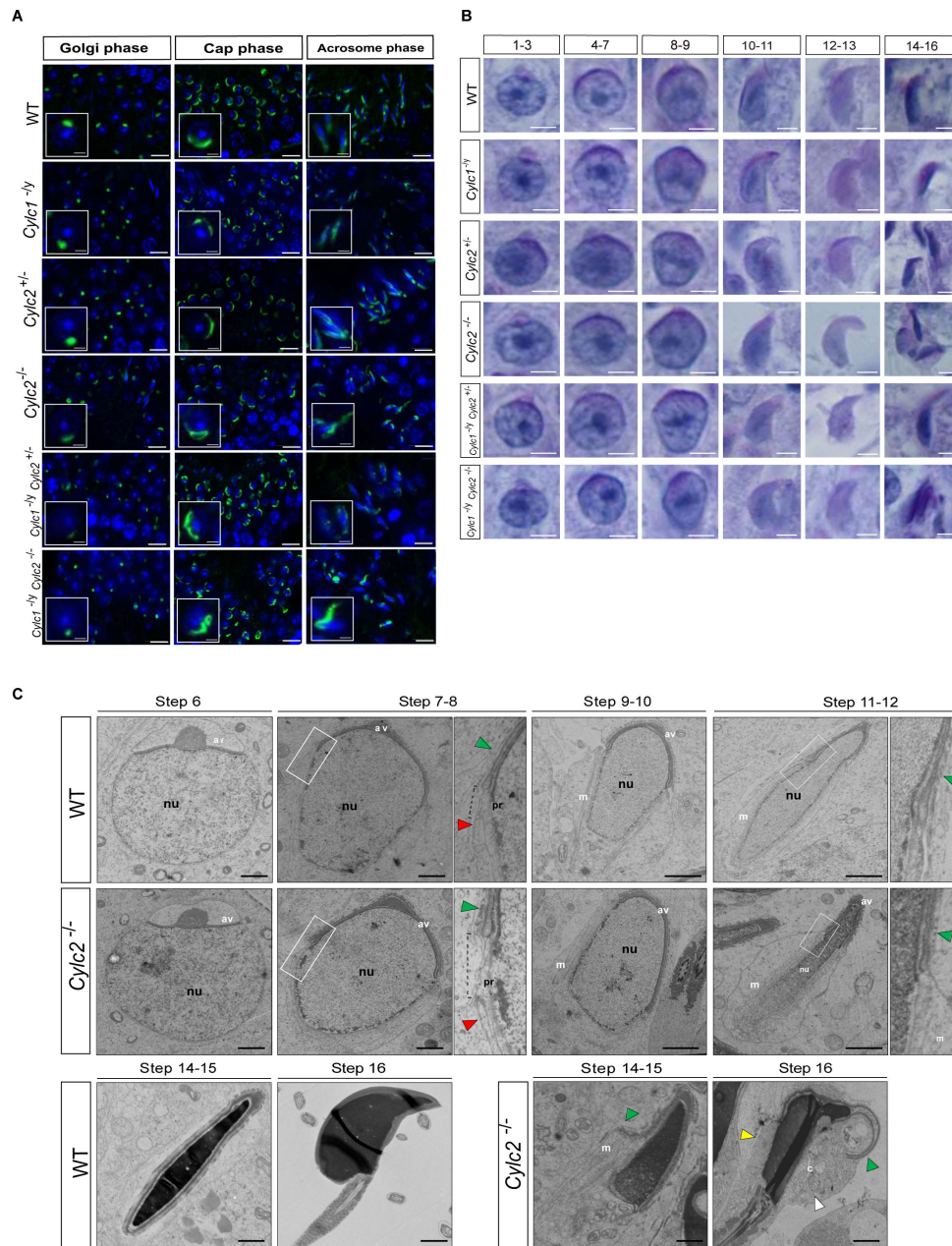


Figure 4

Cylicins are required for acrosome attachment to the nuclear envelope

(A) PNA-FITC lectin immunofluorescence staining of the acrosome in testicular tissue of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} mice (green). Golgi phase of acrosome biogenesis at round spermatid stage (I-IV) is visible in the left panel. Middle panel shows cap phases on round spermatids (stage V-VIII). In the right panel acrosomal phase is shown (stage IX-XI). Nuclei were counterstained with DAPI. Staining was performed on 3 animals from each genotype. Scale bar: 10 μ m. Insets show representative single spermatids at higher magnification (scale bar: 2 μ m).

(B) PAS staining of testicular sections from WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} mice. Representative spermatids at different steps of spermiogenesis are shown. Scale bar: 3 μ m.

(C) TEM micrographs of testicular tissues of WT and *Cylc2*^{-/-} mice. Single spermatids from step 6 to step 16 are shown. nu: nucleus; av: acrosomal vesicle; pr: perinuclear ring; m: manchette microtubules; cy: cytoplasm; green arrowheads: developing acrosome; red arrowheads: manchette; white arrowhead: cytoplasm; yellow arrowhead: remaining microtubules in mature sperm. Scale bar: 1 μ m.

spermiogenesis (**Fig. 5 A** [↗](#)). However, starting from step 12, spermatids from *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-}, and *Cylc1*^{-/-}*Cylc2*^{-/-} mice showed abnormal manchette elongation, which became more prominent at step 13 (**Fig. 5 A** [↗](#)). Manchette length was measured starting from step 10 until step 13 spermatids and the mean was obtained, showing that the average manchette length was 76-80 nm in wildtype, *Cylc1*^{-/-} and *Cylc2*^{+/-}, while for *Cylc2*^{-/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} spermatids mean manchette length reached 100 nm (**Fig. 5 B** [↗](#)). *Cylc1*^{-/-}*Cylc2*^{+/-} spermatids displayed an intermediate phenotype with a mean manchette length of 86 nm. Interestingly, some of *Cylc2*-deficient spermatids showed shifting of the manchette to the ventral side of the nucleus along with excessive elongation. At step 16, the manchette was normally disassembled in wildtype, *Cylc1*^{-/-}, and *Cylc2*^{+/-} spermatids (**Fig. 5 A** [↗](#)). However, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-}, and *Cylc1*^{-/-}*Cylc2*^{-/-} spermatids showed a persistent α -tubulin signal, indicating that disassembly of the manchette is delayed or incomplete (**Fig. 5 A** [↗](#)).

Other than α -tubulin, we also used HOOK1 as manchette marker. HOOK1 is a member of a family of coiled-coil proteins, which bind to microtubules and organelles and regulate microtubule trafficking during endocytosis and spermiogenesis. Co-staining of the spermatids with antibodies against PNA lectin (green) and HOOK1 (red) revealed that abnormal manchette elongation and acrosome anomalies simultaneously occurred in elongating spermatids of *Cylc2*^{-/-} male mice (**Fig. 5 C** [↗](#)). Schematic representation shows acrosome biogenesis and manchette development in wildtype and *Cylc2*^{-/-} spermatids (**Fig. 5 C** [↗](#)). While round spermatids of *Cylc2*^{-/-} mice elongated as those of the wildtype sperm, the manchette elongated abnormally and the acrosome became loosened (**Fig. 4 C** [↗](#), **Fig. 5 C** [↗](#)).

Cylc2 coding sequence is slightly more conserved among species than Cylc1

To address why *Cylc2* deficiency causes more severe phenotypic alterations than *Cylc1* deficiency in mice, we performed evolutionary analysis of both genes. Analysis of the selective constraints on *Cylc1* and *Cylc2* across rodents and primates revealed that both genes' coding sequences are conserved in general, although conservation is weaker in *Cylc1* trending towards a more relaxed constraint (**Fig. 6** [↗](#)). A model allowing for separate calculation of the evolutionary rate for primates and rodents, did not detect a significant difference between the clades, neither for *Cylc1* nor for *Cylc2*, indicating that the sequences are equally conserved in both clades.

To analyze the selective pressure across the coding sequence in more detail, we calculated the evolutionary rates for each codon site across the whole tree. According to the analysis, 34% of codon sites were conserved, 51% under relaxed selective constraint, and 15% positively selected. For *Cylc2*, 47% of codon sites were conserved, 44% under neutral/relaxed constraint, and 9% positively selected. Interestingly, codon sites encoding lysine residues, which are proposed to be of functional importance for Cylicins, are mostly conserved. For *Cylc1*, 17% of lysine residues are significantly conserved and 35% of significantly conserved codons encode for lysine. For *Cylc2*, this pattern is even more pronounced with 27.9% of lysine codons being significantly conserved and 24.3% of all conserved sites encoding for lysine (**Fig. 6** [↗](#)).

Cylicins are required for normal sperm morphology in human

As loss of two Cylicin alleles causes fertility defects in mice, we next addressed whether infertile men also display variants in *CYLC1*/*CYLC2*.

Exome sequencing within the MERGE (Male Reproductive Genomics study) cohort identified one patient (M2270) carrying rare (MAF < 0.01, gnomAD) missense variants in both *CYLC1* and *CYLC2*. The man of German origin presented at age of 40 years for couple infertility because unsuccessfully trying to conceive for 6 years. The couple underwent one ICSI procedure which resulted in 17 fertilized oocytes out of 18 retrieved. Three cryo-single embryo transfers were performed in spontaneous cycles, but no pregnancy was achieved.

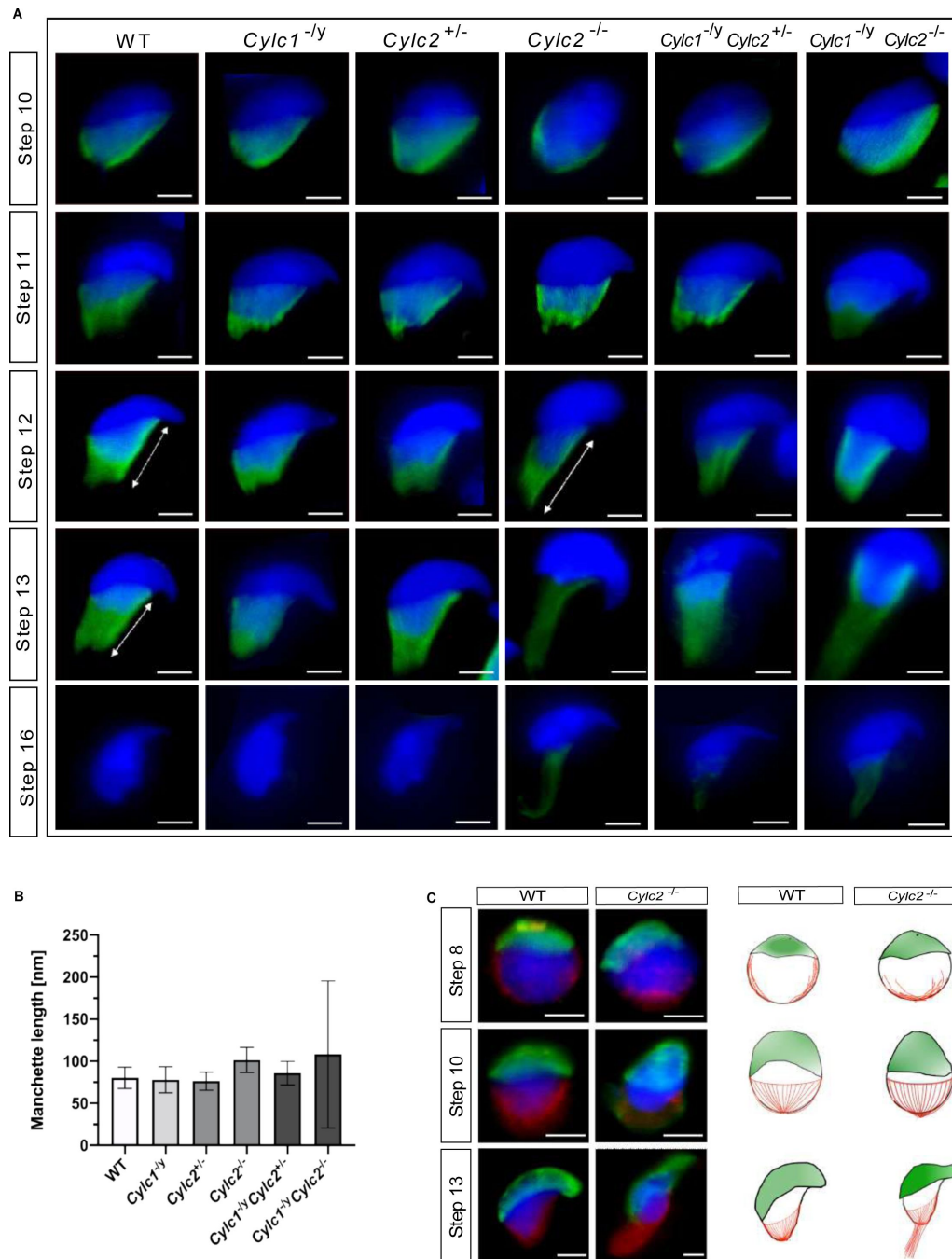


Figure 5

***Cylc2* deficiency causes delay in manchette removal**

(A) Immunofluorescence staining of α -Tubulin to visualize manchette structure in squash testis samples of WT, *Cylc1*^{-ly}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-ly} *Cylc2*^{+/-} and *Cylc1*^{-ly} *Cylc2*^{-/-} mice. Spermatids in different steps of spermiogenesis were shown, for step-to-step comparison. Scale bar: 5 μ m.

(B) Quantification of manchette length in WT, *Cylc1*^{-ly}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-ly} *Cylc2*^{+/-} and *Cylc1*^{-ly} *Cylc2*^{-/-} α -Tubulin stained spermatids at step 10-13.

(C) Co-staining of the manchette with HOOK1 (red) and acrosome with PNA-lectin (green) is shown in round, elongating and elongated spermatids of WT (upper panel) and *Cylc2*^{-/-} mice (lower panel). Scale bar: 3 μ m. Schematic representation shows acrosomal structure (green) and manchette filaments (red).

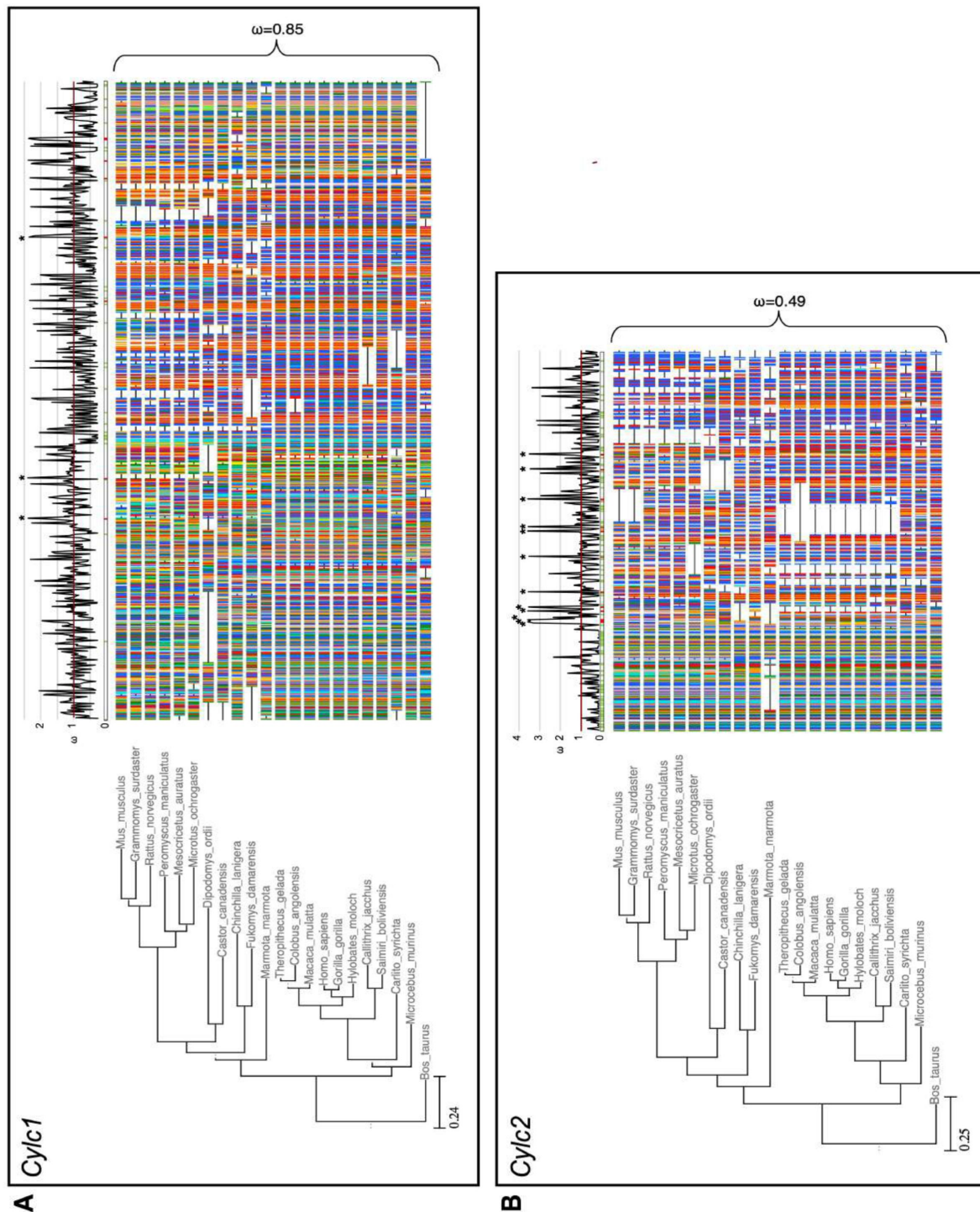


Figure 6

Species phylogeny with branch length representing number of nucleotide substitutions per codon with schematic representation of (A) CYLC1 and (B) CYLC2 amino acid alignment used in the PAML CodeML analysis. Alignments were stripped of columns with gaps in more than 80% of species. Evolutionary rate (ω) obtained by CodeML models M0 is shown for the whole alignment. The graph on top shows the evolutionary rate (ω) per codon sites across the whole tree (CodeML model M2a). Significantly positively selected sites are marked by asterisks. Sites with a probability of higher than 0.95 to belonging to the conserved or positively selected site class are marked in green and red respectively below the graph.

Patient M2270 carries the hemizygous variant c.1720G>C in *CYLC1* that leads to an amino acid exchange from glutamic acid to glutamine (p.(Glu574Gln)), is predicted to be deleterious or possibly damaging by *in silico* tools (SIFT ²⁵ and PolyPhen ²⁶, respectively), and has a CADD score of 11.91. It is located in exon four out of five and affects a region that is predicted to be intolerant to such substitutions (Figure 7 – supplement 1, metadome). It occurs only twice in the gnomAD database (v2.1.1) comprising 141,456 individuals (67,961 XY), once identified in a hemizygous male and once in a female carrier and is absent from our database.

M2270 further carries the heterozygous variant c.551G>A in *CYLC2* that is predicted to be tolerated (SIFT) or benign (PolyPhen) in accordance with a low CADD score of 0.008. It is located in exon five out of eight and affects a region in which variants are likely to be tolerated (Figure 7 – supplement 1, metadome). However, it is a rare variant occurring with an allele frequency of 0.0035 in the general population, according to the gnomAD database. Importantly, only three XX individuals are reported to be homozygous for the variant within *CYLC2*.

Segregation analyses revealed maternal inheritance of the X-linked *CYLC1* variant c.1720G>C p.(Glu574Gln), whereas the father carries the heterozygous *CYLC2* variant c.551G>A p.(Gly184Asp) (**Fig. 7 A**). According to ACMG-AMP criteria ²⁷, both variants are classified as variants of uncertain significance. No other potentially pathogenic variants in genes associated with sperm morphological defects were identified by exploring the exome data of M2270.

Semen analysis performed following WHO guidelines ²⁸ is shown in **Table 1**. The sperm concentration in the semen was slightly reduced, while significant reduction of motile spermatozoa (12.5%) was observed. Interestingly, only 2% of the sperm cells appeared morphologically normal, while 98% of sperm showed head defects (**Table 1**).

Immunofluorescence staining of *CYLC1* revealed that while in healthy donor's sperm, *CYLC1* localizes in the calyx, in M2270, *CYLC1* labeling was absent (**Fig. 7 B**). Bright field microscopy demonstrated that M2270's sperm flagella coiled in a similar manner compared to flagella of sperm from *Cylicin*-deficient mice. Quantification revealed 57% of M2270 sperm displaying abnormal flagella compared to 6% in the healthy donor (**Fig. 7 D**). Interestingly, DAPI staining revealed that 27% of M2270 flagella carry cytoplasmic bodies without nuclei and could be defined as headless spermatozoa (**Fig. 7 C**, white arrowhead; **Fig. 7 E**). CCIN staining demonstrated that while spermatozoa of a healthy donor showed a typical, funnel-like calyx structure in the posterior region of PT, spermatozoa from M2270 had CCIN localized in irregular manner throughout head and tail, suggesting that *Cylicins* have a role in maintenance of the calyx structure and composition in both mice and human spermatozoa (**Fig. 7 F**). Testis-specific phospholipase C zeta 1 (PLCζ1) is localized in the postacrosomal region of PT in mammalian sperm ²⁹ and has a role in generating calcium (Ca²⁺) oscillations that are necessary for oocyte activation ³⁰. Staining of healthy donor's spermatozoa showed a previously described localization of PLCζ1 in the calyx, while sperm from M2270 patient presents signal irregularly through the PT surrounding sperm heads (**Fig. 7 G**). These results suggest that *Cylicin* deficiency can cause severe disruption of PT in human sperm as well, causing male infertility.

Discussion

Spermiogenesis is a highly organized process that is dependent on a unique cytoskeletal organization of the sperm cells. The PT has a role of structural scaffold that surrounds sperm nucleus, and its protein composition is crucial for correct sperm development. In this study, we used CRISPR/Cas9 gene editing to establish *Cylc1*-, *Cylc2*- and *Cylc1/2*-deficient mouse lines to analyze the role of *Cylicins*. We demonstrated that the loss of *Cylicins* impairs male fertility in mice by severely disturbing sperm head architecture. The significance of our findings is supported by the identification of *CYLC1/2* variants in an infertile patient who presents similar structural anomalies in sperm cells.

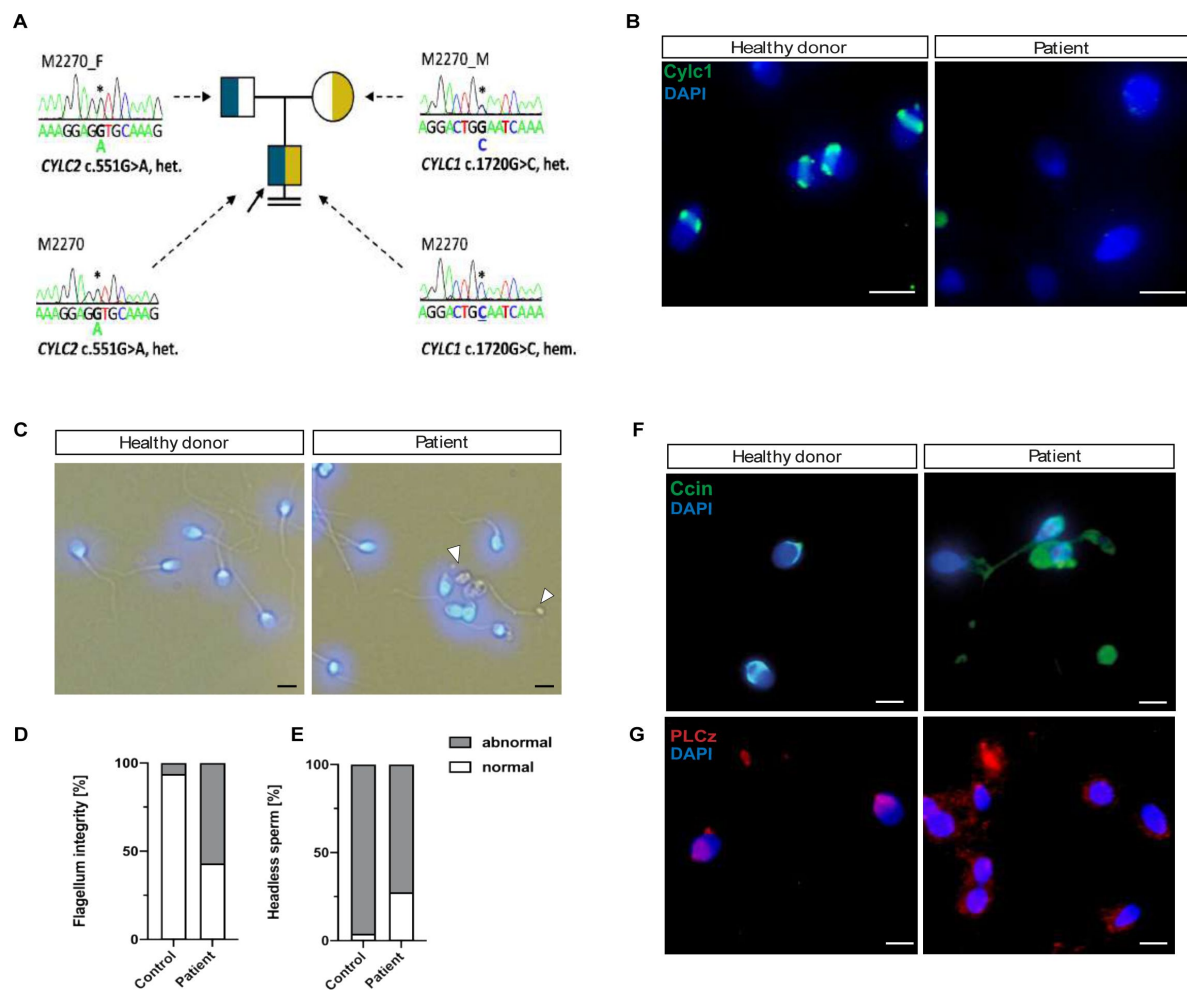


Figure 7

Cylicins are required for human male fertility

(A) Pedigree of patient M2270. His father (M2270_F) is carrier of the heterozygous *CYLC2* variant c.551G>A and his mother (M2270_M) carries the X-linked *CYLC1* variant c.1720G>C in a heterozygous state. Asterisks (*) indicate the location of the variants in *CYLC1* and *CYLC2* within the electropherograms.

(B) Immunofluorescence staining of *CYLC1* in spermatozoa from healthy donor and patient M2270. In donor's sperm cells *CYLC1* localizes in the calyx, while patient's sperm cells are completely missing the signal. Scale bar: 5 μ m.

(C) Bright field images of the spermatozoa from healthy donor and M2270 show visible head and tail anomalies, coiling of the flagellum as well as headless spermatozoa, which carry cytoplasmic residues without nuclei (white arrowhead). Heads were counterstained with DAPI. Scale bar: 5 μ m.

(D-E) Quantification of flagellum integrity (D) and headless sperm (E) in the semen of patient M2270 and a healthy donor.

(F-G) Immunofluorescence staining of CCIN (F) and PLCz (G) in sperm cells of patient M2270 and a healthy donor. Nuclei were counterstained with DAPI. Scale bar: 3 μ m.

	First visit	Second visit	WHO reference rang
Abstinence time [d]	4.0	5.0	
Volume [ml]	4.2	5.8	> 1.4
Concentration [Mill./ml]	10.5	16.3	> 16
Total sperm count [Mill.]	44.1	94.5	> 39
Vitality [%]	53	27	> 54
Motility			
a [%]	7	9	a + b >30
b [%]	5	4	
c [%]	19	8	
d [%]	69	79	
Morphology			
Normal [%]	2	2	> 4
Head defects [%]	99	99	
Midpiece defects [%]	63	59	
Flagella defects [%]	18	47	

Table 1

Semen analysis of the patient M2770 carrying variants in the *CYLC1* and *CYLC2* genes

Detailed analysis of sperm development and morphology demonstrated that infertility of Cylicin-deficient male mice is caused by anomalies of different sperm structures. The prominent loss of PT integrity and calyx structure in the mature sperm lead to acrosome loosening and detachment from the nuclear envelope. Furthermore, the shifting of the basal plate caused damage in the head-tail connecting piece resulting in coiling of the flagellum and impaired swimming. Murine Cylicins are first detected in the periacrosomal region of round spermatids and move to the postacrosomal region of PT during spermatid elongation to finally localize to the calyx of mature sperm. This dynamic localization pattern of Cylicins has been described in human, boar and bovine sperm as well ^{2, 14, 15}.

Interestingly, other PT enriched proteins CCIN ^{12, 31, 7} and CPβ3/CPα3 move in the similar manner across PT during spermiogenesis and are present in the calyx of mature sperm of various mammalian species (Figure 2 – supplement 4). This co-localization suggests the potential interaction between calyx proteins. Furthermore, CCIN ^{2, 12, 13} and CPβ3-CPα3 complex ^{16–18} are described as actin-binding proteins and porcine *Cylc2* has been shown to have a high affinity for F-actin as well ²⁰. The potential roles of F-actin during spermiogenesis in mammals involve biogenesis of the acrosome ³² and its correct attachment to the outer nuclear membrane of the spermatids ³³ as well as removal of excess cytoplasm ³⁴.

The loss of Cylicins caused acrosome detachment from NE starting from cap phase of the acrosome biogenesis. Interestingly, loss of CCIN results in similar loosening of the acroplaxome from the outer nuclear membrane ⁷. CCIN is shown to be necessary for the IAM-PT-NE complex by establishing bidirectional connections with other PT proteins. Zhang et al. found CYLC1 to be among proteins enriched in PT fraction ⁷. Based on their speculation that CCIN is the main organizer of the PT, we hypothesize that both CCIN and Cylicins might interact, either directly or in a complex with other proteins, in order to provide the ‘molecular glue’ necessary for the acrosome anchoring. Furthermore, Cylicin deficiency resulted in cytoplasmic retention and bending of the midpiece, similarly to the CPα3 mutant phenotype repro32/repro32 ³⁵. The CAPZ complex has been shown to regulate the actin dynamics by preventing addition or loss of G-actin subunits to the ends of F-actin filaments ³⁶. Since we demonstrated that the loss of Cylicins resulted in mislocalization of both CCIN and CPα3, we speculate that Cylicins have a crucial role in maintenance of the integrity of PT structure and thus are required for proper function of PT proteins.

Other than morphological defects of the mature sperm PT, during spermiogenesis Cylicin deficiency results in excessive manchette elongation, its delayed disassembly as well as formation of abnormal gaps in the PT at the level of perinuclear ring. The intra-manchette transport (IMT) of proteins from the apical pole of the head to the base of the developing tail occurs during spermiogenesis ³⁷. The malfunctions of the IMT can cause delay in manchette clearance and morphological defects of mature sperm. Mouse models deficient for IMT proteins such as HOOK1 ³⁸, CEP131 ³⁹ and IFT88 ⁴⁰ show abnormal manchette elongation and delay in its clearance resulting in aberrant nuclear shape of the sperm. Furthermore, the loss of HOOK1 results in head-tail connection and mid-piece anomalies such as flipping of the head, basal plate defects and coiling of the tail similar to *Cylc2* deficiency ⁴¹. The localization of manchette on the caudal portion of spermatids coincides with the localization of the calyx in the mature sperm, so these results indicate that the manchette might be maintained longer to compensate for the missing formation of the calyx structure. Furthermore, we observed wide gaps in the perinuclear ring during manchette elongation, suggesting that Cylicins might have a role in maintenance of the contact between caudal and apical PT region and its integrity.

Our evolutionary analysis of *Cylc1* and *Cylc2* genes across rodents and primates revealed that both coding sequences are under purifying selection. Overall, the results reveal that *Cylc1* is under slightly less conserved constraint than *Cylc2* leading to the hypothesis that loss of function in *Cylc1* might be less severe and could be compensated for by *Cylc2* due to partial redundancy. This

hypothesis is supported by our finding that *Cylc1* deficiency causes subfertility in male mice, while the loss of both *Cylc2* alleles results in male infertility. Furthermore, in *Cylc2*^{+/-} male mice fertility was preserved, while *Cylc1*^{-/-}*Cylc2*^{+/-} males were unable to sire offspring leading to the conclusion that the loss of one *Cylc2* allele could be compensated by *Cylc1*, however at least two functional Cylicin alleles are required for male fertility in mice. Interestingly, *Cylc1*^{-/-}*Cylc2*^{+/-} males displayed an “intermediate” phenotype, showing slightly less damaged sperm than *Cylc2*^{-/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} animals. This further supports our notion, that loss of the less conserved *Cylc1* gene might be at least partially compensated by the remaining *Cylc2* allele.

In general, the evolutionary rate of C-terminal lysine rich region of both Cylicins seems to be highly volatile between conserved and positively selected codon sites, while the lysine residues seem to be strongly conserved. Changes in the C-terminal region, potentially affecting the length of the lysine rich domain, might have an adaptive advantage. Targeted positive selection on codon sites could also be a signature of co-evolution with a rapidly evolving interactor.

Sperm morphological defects and infertility observed in one patient with variants in both Cylicin genes points towards a requirement for human spermiogenesis and fertility. A defect in acrosome formation and the existence of variants in two out of three alleles in both *CYLC* genes is in line with the observations made in mice. The absence of *CYLC1* confirmed through immunofluorescent staining indicates a functional impact of the missense variant on the *CYLC1* protein. Furthermore, impaired CCIN localization was observed in patient M2270 as well, suggesting that PT has similar roles in human and rodents despite the differences in sperm head shape ⁴². These results suggest that establishing mouse models deficient for Cylicin and other PT proteins might provide insights into mechanisms of human spermiogenesis and cytoskeletal regulation as well. However, with our data we cannot exclude the possibility that there is a discrepancy between mice and men and that the missense variant in *CYLC1* might alone be sufficient to cause the observed phenotype of M2270. The o/e ratios of both genes calculated within the gnomAD database rather indicate a stronger selective pressure on human *CYLC1* (0.08) than on *CYLC2* (0.84). Furthermore, the *CYLC1* variant affects an intolerant region within the protein sequence according to metadome ⁴³, whereas the *CYLC2* variant is located in a region that is more tolerant to variation (Figure 7 – supplement 1). Therefore, we cannot definitively validate the hypothesis of an oligogenic disease in men as well.

The *CYLC2* missense variant is inherited by the father of M2270 (Fig. 5A) and, thus, not sufficient to cause infertility. However, the father reported difficulties to conceive naturally. Based on this one family, we cannot exclude an effect of pathogenic heterozygous *CYLC2* variants.

Overall, the identification and detailed characterization of further patients with variants in *CYLC1* and *CYLC2* is warranted to draw firm conclusions on the effect of variants in these genes on spermiogenesis and infertility.

Acknowledgements

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Figure 1 – supplement 1: Amino acid sequence comparison of CYLC1 and CYLC2 in *C.elegans* and *M.musculus* to *H.sapiens*. KKD/E motifs are highlighted in blue and repeating units are marked by red brackets.

Figure 1 – supplement 2: Immunohistochemical staining against CYLC1 and CYLC2 in tissue sections of testis, brain, thymus and spleen. Scale bar: 100 μ m.

Figure 1 – supplement 3: Immunofluorescence staining against the acrosomal matrix marker protein SP56 (green) and CYLC1 or CYLC2 (red) in round and elongating spermatids. Nuclei were stained with DAPI. Scale bar: 5 μ m.

Figure 1 – supplement 4: Immunofluorescence staining of CYLC1 and CYLC2 in elongating spermatids of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} mice. Scale bar: 5 μ m.

Figure 1 – supplement 5: Proteome abundances

Figure 1 – supplement 6: Proteome clustering

Figure 2 – supplement 1: HE-stained testicular tissue sections of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} mice. Scale bar: 100 μ m.

Figure 2 – supplement 2: Eosin-Nigrosin staining of epididymal sperm samples from WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} mice. Scale bar: 10 μ m.

Figure 2 – supplement 3: Nuclei of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} sperm stained with DAPI. Scale bar: 5 μ m. Elongation and circularity of nuclei from WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} sperm. The minimum detection area was set to 1.000 pixels, while the maximum detection area was 7.000 pixels.

Figure 2 – supplement 4: Co-staining against CYLC1/CYLC2 (red) and CCIN (green) in epididymal sperm cells of WT mouse. Nuclei were counterstained with DAPI. Scale bar: 2 μ m.

Figure 3 – supplement 1: TEM micrographs of WT and *Cylc2*^{-/-} sperm and axonemes.

Figure 3 – supplement 2: SpermQ analysis of the flagellar beat of WT (green) and *Cylc2*^{-/-} (red) sperm. Average curvature of the flagellum and the arc length are shown.

Figure 4 – supplement 1: PNA-lectin immunofluorescence staining of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} testicular tissue. Spermatids at cap phase and acrosome phase of acrosome biogenesis are shown individually. Scale bar: 5 μ m.

Figure 4 – supplement 2: PAS stained testicular tissue sections of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} mice. Scale bar: 20 μ m.

Figure 4 – supplement 3: TEM micrographs of degrading damaged spermatids in testicular sections of *Cylc2*^{-/-} mice. Scale bar: 5 μ m.

Figure 5 – supplement 1: Immunofluorescence staining of α -Tubulin in WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-} squash testis samples. Spermatids at step 8-9 are shown. Scale bar: 10 μ m.

Figure 7 – supplement 1:

(A) Localization of the CYLC1 variant found in M2270. The variant affects exon four and an intolerant part of the C-terminal region of CYLC1 according to metadome 41.

(B) CYLC2 variant localization. The missense variant in CYLC2 detected in M2270 affects exon five and a tolerant part of CYLC2 according to metadome 41.

Methods

Animals

All animal experiments were conducted according to German law of animal protection and in agreement with the approval of the local institutional animal care committees (Landesamt für Natur, Umwelt und Verbraucherschutz, North Rhine-Westphalia, approval IDs: AZ84-02.04.2013.A429, AZ81-02.04.2018.A369). *Cylc1*- and *Cylc2*-deficient mice were generated by CRISPR/Cas9-mediated gene-editing in zygotes of the hybrid strain B6D2F1. Guide sequences are depicted in [Table 2](#). For *Cylc1*, in vitro transcribed sgRNAs (50 ng/μl each) and Cas9 mRNA (100 ng/μl) (Sigma-Aldrich, Taufkirchen, Germany) were microinjected into the cytoplasm of zygotes as described previously⁴⁴. *Cylc2*-deficient mice were generated by electroporation of ribonucleoprotein (RNP) complexes into zygotes using a GenePulser II electroporation device (BioRad, Feldkirchen, Germany). For RNP formation, crRNA and tracrRNA (IDT, Leuven, Belgium) were combined in duplex buffer (IDT) to a final concentration of 50 mM each and annealed (95°C, 5 min; cool down to room temperature with -0.2°C/sec.). For RNP assembly, 4 μM Cas9 protein (IDT) and 2 μM of each annealed cr/tracr RNA were combined in Opti-MEM media (Gibco 11058-021, Thermo Fisher, Carlsbad, USA) and incubated for 10 min at room temperature. RNPs complexes were diluted 1:2 in Opti-MEM, supplemented with 30-40 zygotes in a 0.1 cm gene-pulser cuvette (BioRad) and electroporated (two 30 V square wave pulses, 2 ms pulse length, 100 ms pulse interval). Recovered embryos were cultured over night at 37°C, 5% CO₂ in G-TL medium (Vitrolife, Göteborg, Sweden) and transferred into the oviduct of pseudopregnant CB6F1 foster mice at 2-cell stage. Offspring were genotyped and gene-edited alleles were separated by backcrossing of founder animals with C57Bl/6J mice. Established mouse lines were registered with Mouse Genome Informatics: *Cylc1*^{em1Hsc} (MGI: 7341368), *Cylc2*^{em1Hsc} (MGI:6718280). Mouse lines were maintained as congenic strains on C57Bl/6J background. For all analyses sexually mature males at the age of 8-15 weeks, backcross generation ≥N3 were used.

Genomic DNA extraction and genotyping

Genomic DNA was extracted from biopsies using the HotShot method⁴⁵. PCR reactions were assembled according to the manufacturers protocol of the DreamTaq Green DNA Polymerase (Thermo Fisher, EP0712) using gene-specific primers listed in [Table 3](#).

Fertility analysis

Males were mated 1:2/1:1 with C57Bl/6J females, which were checked daily for presence of a vaginal plug indicative for successful copulation. Plug positive females were separated and pregnancies as well as litter size were recorded.

Sampling

Mature sperm were obtained from the cauda epididymis in M2 medium (Merck) or PBS preheated at 37°C. Following several incisions of the cauda, sperm were retrieved by flush-out for 15-30 min. The extracted sperm samples were washed in PBS, centrifuged (4000 rpm, 5 min, 4°C) and snap

Name	protospacer sequence (5'-3')
<i>Mm.Cas9.CYLC1.sg1</i>	GGTTTATCCATACGTGAGT
<i>Mm.Cas9.CYLC1.sg2</i>	GGCTTAGGTGATGCTCTAAA
<i>Mm.Cas9.CYLC2.1.AB</i>	AAGGGAGAGTCGAAAAGCGT
<i>Mm.Cas9.CYLC2.1.AF</i>	GGATCCAAGGATAAAGTGTC

Table 2

Protospacer sequences

Cylc1	5'-3'	expected band size
<i>Cylc1_F1</i>	TATACACACAATCCACAATCTTGAAAT	WT: 427 bp
<i>Cylc1_R1</i>	TCACTTCAAAATCCAACTTGCCT	KO: 264 bp
<i>Cylc1_R2</i>	TGCCTAGTATTCAGGTTCCCC	
Cylc2		
<i>Cylc2_F1</i>	ACCACCATTATGGATGCACCG	WT: 376 bp
<i>Cylc2_R1</i>	AGTGTTTCTTGAGCTCGTTG	KO: 286 bp
<i>Cylc2_R2</i>	GGCTGAATCTTACCCTTAGGT	

Table 3

PCR primer sequences

frozen. For squash tubule samples, fresh testes from three animals of each genotype were transferred to PBS and tubules were dissected as described by Kotaja et al.⁴⁶. Tubule sections were pressed on a slide, quickly frozen in liquid nitrogen and fixed in 90% ethanol for 5 min.

Semen analysis

Sperm concentrations for at least 4 animals of each genotype were determined using a Neubauer hemocytometer. Viability of sperm was determined by Eosin-Nigrosin staining as described previously⁴⁷ for at least 3 animals per genotype. Sperm motility was analyzed referring to the WHO guidelines for analysis of human semen. For all analyses at least 100 sperm per individual were analyzed and the percentage of motile/immotile, viable/inviable sperm was calculated.

Quantitative reverse transcription-polymerase chain reaction (qRT-PCR)

RNA was extracted from the testis tissue with TRIzol (Life Technologies, Carlsbad, USA; 15596018). RNA concentrations and purity ratios were determined by NanoDrop ONE (Thermo Scientific) measurements. qRT-PCR was performed as described previously⁴², on ViiA 7 Real Time PCR System (Applied Biosystems) using Maxima SYBR Green qPCR Mastermix (Thermo Fisher; K0221). Replicate of 3 was used for each genotype. Beta-actin was used as a housekeeping gene for normalization. Primers used are shown in [table 4](#).

Subcellular protein extraction and Western Blot analysis

For the extraction of cytoskeletal protein fraction from sperm cells, Subcellular Fractionation Kit for Cultured cells (Thermo Fisher, #78840) was used according to manufacturer's instructions, with slight modifications to adjust it to sperm cells. Five subcellular protein fractions were extracted: membrane protein fraction, cytoplasmic protein fraction, soluble nuclear fraction, chromatin bound fraction and cytoskeletal protein fraction. Briefly, epididymal sperm cells were washed in ice cold PBS and centrifuged at 500 x g for 5 min at 4°C. Pellet was resuspended in Cytoplasmatic Extraction Buffer (CEB), incubated for 10 min and centrifuged again at 500 x g for 5 min. Following steps of subcellular fractionation were performed according to manufacturer's instructions. Quantity of proteins in each fraction was determined using NanoDrop ONE (Thermo Fisher). For further analysis only cytoskeletal protein fraction was used.

Cytoskeletal proteins were separated on 12% SDS gel with 5% stacking gel. Transfer to PVDF membrane was performed using Trans Blot Turbo System (Bio-Rad). Membranes were washed with TBST 1X, stained with Coomassie blue and blocked with 1% milk for 1 h at room temperature with gentle shaking. Primary antibodies anti-CYLC1 and anti-CYLC2 were diluted in milk and incubated overnight at 4°C (for antibody dilutions see [table 5](#)). After washing in TBST, membranes were incubated for 1 h at room temperature with polyclonal goat anti-rabbit secondary antibody IgG/HRP (P044801-2; Agilent Technologies/Dako, Santa Clara, 527 CA, United States) diluted 1:2000 in milk. After extensive TBST washing membranes were imaged using WESTARNOVA2.0 chemiluminescent substrate (Cyanagen) or SuperSignal West Femto Maximum Sensitivity Substrate (34095; Thermo Fisher) and ChemiDoc MP Imaging system (Bio-Rad). Membranes were further re-blocked in 3% BSA for 1 h at room temperature with gentle shaking and incubated with α -Tubulin at 4°C overnight. After washing in TBST, membranes were incubated for 1 h at room temperature with polyclonal rabbit anti-mouse secondary antibody IgG/HRP (P0260; Agilent Technologies/Dako) diluted 1:2000 in 3% BSA.

Name	forward (5'-3')	reverse (5'-3')
<i>Cylc1</i>	GGGGAAAAATAAGCTCATGTGTAG	TTCAGGTTCCCCATTGGTTA
<i>Cylc2</i>	GCATTTCCTCAAACCAAGG	AACGGATGGTCTCTCGGATATT
<i>beta-actin</i>	TGTTACCAACTGGGACGACA	GGGTGTTGAAGGTCTCAA

Table 4

qRT primer sequences

Antibody	company	Catalogue number	Antigen	Dilution IF	Dilution WB
α -Tubulin	Abcam	ab7291			1:10000 in 3% BSA
α -Tubulin	Merck Millipore (Billerica, USA)	16-232		1:1000	
CapZ α 3	Progen	GP-SH4		1:500	
Ccin	Progen	GP-SH3		1:500	
Cylc1 (used for mouse)	Davids Biotechnology (Regensburg, Germany)	Custom made polyclonal antibody	AESRKSNDERRKTLKIKFRGK and KDAKKEGKKKGKRESRKKR	1:1000 (sperm cells) 1:500 (testis tissue) 1:1000 Western blot	1:1000 in 5% milk in TBST
Cylc1 (used for human samples)	Santa Cruz	sc-166400		1:500	
Cylc2	Davids Biotechnology (Regensburg, Germany)	Custom made polyclonal antibody	KSVGTHKSLASEKTKKEVK and ESGGEKAGSKKEAKDDKKDA	1:1000 (sperm cells) 1:500 (testis tissue) 1:1000 Western blot	1:1000 in 5% milk in TBST
Hook1	Proteintech	10871-1-AP		1:500	
PLC ζ	Invitrogen	PA5-50476		1:100	
Sp56	Invitrogen	MA1-10866		1:500	

Table 5

Antibodies

Proteomics

Peptide preparation

All chemicals from Sigma unless otherwise noted.

Cytoskeletal protein solutions extracted as described in previous paragraph were processed with the SP3-approach ⁴⁸. Briefly, protein lysate with 50 µg protein were subjected to cysteine reduction and alkylation with 20 mM DTT and 40 mM acrylamide in 50 mM triethylammonium bicarbonate (TEAB). Then a mixture of hydrophilic carboxylate-coated magnetic beads (equal amounts of Sera-Mag SpeedBeads, GE Healthcare, cat. no. 45152105050250, and cat. no. 65152105050250) were added at a bead:protein ratio of 10:1 (w/w). Protein binding was induced by adding 3 volumes of ethanol and subsequent mixing for 5 min. Beads with bound proteins were then washed three times with 80% ethanol and finally subjected to overnight tryptic digestion at 37 °C using a trypsin:protein ratio of 1:25. Peptide solutions were separated from the magnetic beads, dried in a vacuum concentrator, and stored at -20°C. Before measurements, 10 µg of peptides were further desalted with C18 ZipTips (Merck Millipore, Darmstadt, Germany) to ensure complete removal of beads.

LC-MS analysis

Dried peptides were dissolved in 10 µl 0.1% formic acid (solvent A) Peptide separation was performed on a Dionex Ultimate nano HPLC system (Dionex GmbH, Idstein, Germany) coupled to an Orbitrap Fusion Lumos mass spectrometer (Thermo Fisher Scientific, Bremen, Germany). 1 µg peptides were injected onto a C18 analytical column (400 mm length, 100 µm inner diameter, ReproSil-Pur 120 C18-AQ, 3 µm).

The samples were analyzed by a standard data-dependent (DDA) method: Peptides were separated during a linear gradient from 5% to 35% solvent B (90% acetonitrile, 0.1% FA) at 300 nL/min within 120 min. Data-dependent acquisition was performed on ions between 330 and 1600 m/z scanned in the Orbitrap detector every 2.5 seconds ($R=120,000$, standard gain control and inject time settings). Polysiloxane (m/z 445.12002) was used for internal calibration. $z>1$ ions were subjected to higher-energy collision induced dissociation (HCD: 1.0 Da quadrupole isolation, threshold intensity 25,000, collision energy 28%) and fragments analyzed in the Orbitrap ($R=15,000$). Fragmented precursor ions were excluded from repeated analysis for 25 s.

Data analysis

Raw data processing of DDA data and analysis of database searches were performed with Proteome Discoverer software 2.5.0.400 (Thermo Fisher Scientific). Peptide identification was done with an in-house Mascot server version 2.8.1 (Matrix Science Ltd, London, UK) against the Uniprot reference proteome for *Mus musculus* (as of 06/28/23) and a collection of common contaminants ⁴⁹. Precursor ion m/z tolerance was 10 ppm, fragment ion tolerance 20 ppm. Tryptic peptides (Trypsin/P) with up to two missed cleavages were searched, propionamide was set as a static modification of cysteines, while oxidation of methionine and acetylation of protein N-termini were set as dynamic modifications. Spectrum confidence of Mascot results was assessed by the Percolator algorithm 3.05 as implemented in Proteome Discoverer ⁵⁰. Spectra without high confident matches (q -value > 0.01) were sent to a second round Mascot search with semi-specific enzyme cleavage and changing the modification of cysteines with propionamide to dynamic. Proteins with two unique proteins in the protein group were reported. For quantification, summed abundances were normalized on total protein amount in Proteome Discoverer.

High-resolution microscopy of the flagellar beat

Image sequences of mouse sperm were acquired using dark-field at an inverted microscope (IX71; Olympus, Hamburg, Germany), equipped with a dark-field condenser and a high-speed camera (ORCA-Flash4.0 V3, C13220-20 CU, Hamamatsu, Hamamatsu City, Japan). A 10x objective (NA 0.4, UPlanFL; Olympus, Hamburg, Germany) with an additional 1.6x magnifying lens (Olympus, Hamburg, Germany) that was inserted into the light path (final magnification: 16x) was applied. Image sequences were recorded at a rate of 200 frames per second (fps). A custom-made observation chamber was used [22](#). Sperm samples were diluted in THY buffer shortly before insertion of the suspension into the observation chamber. Three WT and three *Cylc2*^{-/-} animals were used.

Sperm nuclear morphology

For the analysis of sperm nuclear morphology, epididymal sperm samples from 3 animals of each genotype were fixed in methanol and acetic acid (3:1). The samples were spread onto a slide and stained with 4',6-diamidino-2-phenylindole (DAPI) containing mounting medium (ROTImount FluorCare DAPI (Carl Roth GmbH, Karlsruhe, Germany; HP20.1)). The sperm cells were imaged at 100x magnification, using a Leica DM5500 B fluorescent microscope. At least 200 pictures were taken from each group and analyzed using Nuclear Morphology software [51](#) according to the developer's instructions. The minimum detection area was set to 1.000 pixels while the maximum detection area was 7.000 pixels.

Histology

Bouin fixed testis tissues were paraffinized, embedded and sectioned at 3-5 µm using microtome. For histological analysis, the sections were deparaffinized, hydrated incubated with Periodic acid (0.5%) for 10 min, rinsed with H₂O and incubated for 20 min with Schiff reagent. Alternatively, tissue sections were stained with Hemalum solution acid (Henricks and Mayer) and Eosin Y solution (Carl Roth). In both procedures, stained sections were dehydrated in alcohol row and mounted using Entellan (Sigma-Aldrich). Slides were imaged at 20x and 63x magnification under bright field using 5500 B microscope.

Immunofluorescence/Immunohistochemistry

Bouin fixed testis tissue sections were deparaffinized in Xylene and rehydrated in decreasing alcohol to be used for immunohistochemical staining. Squash testis samples fixed in 90% EtOH were used for the staining of α-Tubulin and HOOK1. For sperm immunofluorescence, mature sperm cells isolated from cauda epididymis were fixed with methanol acetic acid (3:1), dropped on glass slides and air dried. After washing in PBS twice, all samples were permeabilized using 0.1% Triton x100 for 10 min at room temperature. The samples were then blocked with 5% BSA for 30 min, followed blocking with normal horse serum (Vectorlabs, Burlingame, USA; DI-1788) for 30 min at room temperature. For tissue sections, heat-activated antigen retrieval was performed using citrate buffer (pH 6.0). All primary antibodies were incubated over night at 4°C. Antibodies and dilutions are listed in [table 5](#). The respective secondary antibodies were incubated for 1 h at room temperature using VectaFluor Labeling Kit DyLight 488 and DyLight 594 (Vectorlabs, Burlingame, USA; DI-1788, DI-1794). Slides were mounted with DAPI containing mounting medium (ROTImount FluorCare DAPI, Carl Roth; HP20.1).

For IHC staining against CYLC1 and CYLC2, after antigen retrieval and blocking procedures, slides were treated with 6% H₂O₂ for 30 min. Slides were then incubated with primary antibodies overnight at 4°C. Biotinylated goat anti-rabbit IgG was used as a secondary antibody and incubated for 1 h at room temperature. Slides were then processed using Vectastain ABC-AP Kit (Vector Laboratories, AK-5001) and stained using ImmPACT® Vector® Red Substrate Kit, Alkaline Phosphatase (Vector Laboratories, SK-5105) according to manufacturer's instructions. Counterstaining was performed using Hematoxylin.

For the analysis of acrosome biogenesis peanut agglutinin (PNA)-fluorescein isothiocyanate (FITC) Alexa Fluor 488 conjugate (Molecular Probes, Invitrogen, Waltham, USA; L21409) was used on the Buin fixed testis tissues. After permeabilization and blocking, the tissues were incubated with PNA-FITC 5 µg/ml for 30 min at room temperature. Mature sperm were fixed with paraformaldehyde (4%) for 20 min at room temperature. After PBS washing, the sperm samples were incubated with 5 µg/ml PNA-FITC and 5 nM Mito Tracker Red (Cell Signaling; 9082) for 30 min at room temperature. The slides were then mounted with DAPI mount. All stainings were performed on minimum of 3 animals per genotype.

Transmission electron microscopy

For Transmission Electron Microscopy fresh epididymal sperm and testis tissue were used. After washing with PBS, the samples were fixed in 3% glutaraldehyde at 4°C overnight. The samples were then washed in 0.1 M cacodylate buffer and fixed again in 2% osmium tetroxide at 4°C for 2 h. After dehydration in an ascending ethanol row, the samples were contrasted in 70% ethanol 0.5% uranyl acetate for 1.5 h at 4°C. Samples were then washed in propyleneoxide, three times for 10 min at room temperature before embedding in Epon C at 70°C for 48 h. Ultra-thin sections were examined using a Verios 460L microscope (FEI) with a STEM III-detector.

Evolutionary analysis

Evolutionary rates of mammalian Cylicin genes were analyzed according to Lüke et al (2016) ⁵²[52](#). Briefly, *Cylc1* and *Cylc2* nucleotide sequences were obtained from NCBI genbank and Ensembl genome browser. Phylogenetic trees of considered species were built according to the “Tree of Life web project”. The webPRANK software was applied for codon-based alignment of orthologous gene sequences and results were visualized using the ETE toolkit. To determine evolutionary rates of gene sequences across mammals, for different clades and for individual codon sites, the codeml application implemented in the PAML software was used ⁵³[53](#), ⁵⁴[54](#). Selective pressures on protein level are represented by calculation of the nonsynonymous/synonymous substitution rate ratio ($\omega = dN/dS$). It distinguishes between purifying selection ($\omega < 1$), neutral evolution ($\omega = 1$) and positive selection ($\omega > 1$) within various models. The M0 model served as basis for all performed analyses. Different codon frequency settings were tested for the M0 model of each gene and the setting with the highest likelihood was chosen. To test whether alternative models describe the selective constraints within a dataset more precise than the M0 model, likelihood ratio-tests (LRT) were performed. Applied models and LRTs are described by Yang ⁵³[53](#), ⁵⁴[54](#) and Lüke et al. ⁵²[52](#).

Study cohort and ethical approval

The MERGE (Male Reproductive Genomics study) cohort currently comprises over 2,030 men, mainly recruited at the Centre of Reproductive Medicine and Andrology (CeRA) in Münster. The large majority has severe spermatogenic failure, i.e. severe oligozoospermia (<5 Mill./ml sperm concentration), crypto- or azoospermia. So far, only 35 cases were included because of notable sperm morphological defects (≥ 5 Mill./ml sperm concentration, <4 % normal forms). Common causes for infertility such as oncologic diseases, AZF deletions or chromosomal aberrations were ruled out in advance. Patients with aetiologically still unexplained infertility underwent whole exome sequencing.

All participants gave written informed consent according to the protocols approved by the Ethics Committee of the Medical Faculty Münster (Ref. No. MERGE: 2010-578-f-S) in accordance with the Declaration of Helsinki in 1975.

Whole exome sequencing and data analysis

After DNA extraction from patients’ peripheral blood lymphocytes WES was performed as previously described ⁵⁵[55](#). WES data obtained from 2,066 infertile men was filtered for rare (≤ 0.01 minor allele frequency, gnomAD) variants located within the coding sequence or the adjacent 15

bp of flanking introns in *CYLC1* and *CYLC2*. Patients carrying only one variant in either of the gene were excluded. In case of either bi-allelic *CYLC2* variants or a combination of *CYLC1* and *CYLC2* variants, the whole exome dataset was screened to rule out other potential genetic causes. Variants detected in this study were classified according to the guidelines by the American College of Medical Genetics and Genomics-Association for Molecular Pathology (ACMG-AMP)²⁷ adapted to recent recommendations as outlined in⁵⁶.

To rule out an alternative genetic cause for the patient's condition, his exome data was screened for rare (MAF ≤ 0.01 , gnomAD and in-house database), homozygous, X-linked or potentially compound heterozygous high-impact variants (stop gain, start lost, stop lost, frameshift, splice site and splice region as well as missense variants with CADD > 20) and rare, heterozygous LoF variants (stop gain, start lost, stop lost, frameshift, splice site) without filtering for a specific set of genes. Respective genes were screened for testis-expression and reports of infertility.

Sanger sequencing

Sanger sequencing with variant-specific primers was conducted for validation and segregation purposes. The primers to confirm the *CYLC1* (NM_021118.3) variant c.1720G>C p.(Glu574Gln) are 5'-ACTGATGCTGACTCTGAACCG-3' (forward) and 5'-CCTTCGAGTCACAAAGTTGTCA-3' (reverse). To confirm the *CYLC2* (NM_001340.5) variant c.551G>A p.(Gly184Asp), the primers 5'-CTGTCGAGGTGGATTCTAAAGC-3' (forward) and 5'-TGATCCTTCTTGCATCCT-3' (reverse) were used.

Analysis of the human sperm samples

Human ejaculate samples from healthy donor and patient M2270, were analyzed according to WHO guidelines prior to washing in buffer and centrifuged (1000 rpm, 20 min). The cells were fixed in methanol and acetic acid (3:1) and used for immunofluorescence staining. Samples were dropped on slides and permeabilized with 0.1% Triton x100. After blocking with 5% BSA for 30 min, slides were incubated with primary CCIN or *CYLC1* antibodies (concentrations shown in **Table 3**) for 3 h at room temperature. Secondary antibodies were incubated for 1 h, followed by mounting with DAPI containing medium. All stainings were repeated 3 times using aliquots of the same sample.

Statistics

For all analyses mean values $\pm$ SD were calculated. Statistical significance was determined by One-way ANNOVA analyses using Bonferroni correction. All experiments were conducted as biological replicates and N is provided in Materials and methods section and/or figure legends.

Materials and data availability statement

Custom made antibodies used in this study were developed against murine *CYLC1* and *CYLC2*. A limited amount of antibody can be provided upon request.

SpermQ software used for the analysis of the flagellar beat is publicly available <https://github.com/hansenjn/SpermQ>.

For the purposes of this study a new code was written for color-coded figures and it's freely available at <https://github.com/hansenjn/ColorStackByTimeAndProject>.

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Reviewer #1 (Public Review):

Mice and humans have two Cylicin genes (X-linked Cylicin 1 and the autosomal Cylicin 2) that encode cytoskeletal proteins. Cylicins are localized in the acrosomal region of round spermatids, yet they resemble a calyx component within the perinuclear theca of mature

sperm nuclei. The function of Cylicins during this developmental stage of spermiogenesis (tail formation and head elongation/shaping) was not known. In this study, using CRISPR/Cas genome editing, the authors generated *Cylc1*- and *Cylc2*-knockout mouse lines to study the loss-of-function of each Cylicin or all together.

The major strengths of the study are the rigorous and comparative phenotypic analyses of all the combinatorial genotypes from the cross between the two mouse lines (*Cylc1*^{-/y}, *Cylc2*^{-/-}, *Cylc1*^{-/y} *Cylc2*^{+/-} and *Cylc1*^{-/y} *Cylc2*^{-/-}) at the levels of male fertility, cellular, and subcellular levels to support the conclusion of the study. While spermatogenesis appeared undisturbed, with germ cells of all types detected in the testis, low sperm counts in epididymis were observed. Mice were subfertile or infertile in a dose-dependent manner where fewer functional alleles had more severe phenotypes; the loss of *Cylc2* was less tolerated than the loss of *Cylc1*. Thus, loss of *Cylc1*, and to an even greater extent, loss of *Cylc2*, leads to sperm structure anomalies and decreased sperm motility. Particularly, the sperm head and sperm head-neck region are affected, with calyx not forming in the absence of Cylicins, the acrosomal region being attached more loosely, and the sperm head itself appearing structurally rounder and shorter. Furthermore, manchette, which disassembles during spermiogenesis, persists in mature sperm of mice missing *Cylc2*. It is interesting that the study identifies a human male that has mutations in both *CYLC1* and *CYLC2* genes and suffers from infertility, with similar motility and sperm structure defects compared to the mouse models. *CYLC1* in the sperm from the infertile patient is absent, providing evidence that in both rodents and primates, Cylicins are essential for male fertility. Evolutionary analysis of two genes adds an interesting point. The authors show that the reason for the loss of *Cylc2* being more severe is due to the higher conservation of *Cylc2* compared to *Cylc1* in rodents and primates.

Overall, the work highlights the relevance and importance of Cylicins in male infertility and advances our understanding of perinuclear theca formation during spermiogenesis.

Reviewer #2 (Public Review):

The work presented in this manuscript focuses on the role of Cylicins in spermiogenesis and the consequences of their absence on infertility. The manuscript is presented in two parts: the first part studies the absence of Cylicins from KO mouse models and shows in mice that both isoforms of Cylicins are necessary for normal spermiogenesis. The evaluation of double heterozygotes is particularly useful for the second part which looks at the presence of mutations in these genes in a cohort of infertile men. A patient with two hemizygous/heterozygous mutations in the *CYLC1* and 2 genes, respectively, was identified for the first time and the results obtained with the KO models support the hypothesis of the pathogenicity of the mutations.

In general, the experiments are perfectly performed and the results are clear. Numerous techniques in the state of the art in male reproduction are used to obtain high-quality phenotyping of the mouse models.

The discovery of two concomitant mutations in an infertile patient is very interesting and the work carried out on mice allows supporting that an absence of *CYLC1* and a heterozygous mutation of *CYLC2* could lead to a phenotype of complete infertility. However, as the mutation on *CYLC2* is not identified as pathogenic, the pathogenicity of this mutation remains in question (the authors note this point in the discussion). It would be interesting to see if the mutated amino acid is conserved between different species. In mice, the authors have shown the importance of these proteins on the morphology of the acrosome. What about in humans?

Reviewer #3 (Public Review):

The authors tried to study the role of the cylicin gene in sperm formation and male fertility. They used the Crispr/cas 9 to knockout two mouse cylicin genes, cylicin 1 and cylicin 2. They used comprehensive methods to phenotype the mouse models and discovered that the two genes, particularly cylicin 2 are essential for sperm calyx formation. They further compared the evolution of the two genes. Finally, they identified mutations of the genes in a patient. The major strengths are the high quality of data presented, and the conclusion is supported by their findings from the animal models and patients. The major weakness is that the study is rather descriptive without molecular mechanism studies, limiting its impact on the field.

Author Response

The following is the authors' response to the original reviews.

We appreciate very much the comments and suggestions on our manuscript "Cylicins are a structural component of the sperm calyx being indispensable for male fertility in mice and human". According to the comments, we performed a series of further experiments, re-worded and re-wrote several paragraphs and re-structured the manuscript according to the reviewers' comment. We think that the manuscript is now improved and are looking forward to the further evaluations. We provide a point by point response to all comments and have prepared a version.

Recommendations for the authors:

Editor's comment:

- 1. As pointed out by all three reviewers, it is critical to show the specificity of the antibodies used. The authors should clarify how the specificity of antibodies is tested. Western blot analysis to show the absence of the protein in the knockout is essential.*

As suggested by all reviewers, we additionally performed Western Blot analysis on cytoskeletal protein fractions to further verify the specificity of generated antibodies and the generation of functional knockout alleles. Results nicely confirm the results of the IF staining, however, both anti-bodies detected the bands lower than the predicted molecular weight. In addition, Mass Spectrometry was performed to search for the presence of peptides in the cytoskeletal protein fractions. The paragraph reads now as follows:

Line 127-134: Additionally, Western Blot analyses confirmed the absence of CYLC1 and CYLC2 in cytoskeletal protein fractions of the respective knockout (Fig. 1 G), thereby demonstrating i) specificity of the antibodies and ii) validating the gene knockout. Of note, the CYLC1 antibody detects a double band at 40-45 KDa. This is smaller than the predicted size of 74 KDa as, but both bands were absent in *Cylc1*^{-/-}. Similarly, the CYLC2 Antibody detected a double band at 38-40 KDa instead of 66 KDa. Again, both bands were absent in *Cylc2*^{-/-}. Next, Mass spectrometry analysis of cytoskeletal protein fraction of mature spermatozoa was performed detecting both proteins in WT but not in the respective knockout samples (Figure 1 – supplement 5; Figure 1 – supplement 6).

Specificity of antibodies was additionally proven by immunohistochemical staining, showing a specific staining only in testis sections but not in any other organ tested. The section reads now as follows:

Line 115-117: Specificity of antibodies was proven by immunohistochemical stainings (IHC), showing a specific signal in testis sections only, but not in any other organ tested (Figure 1 – supplement 2)

1. Re-structuring/streamlining of the figures is recommended. Please consider the flow suggested by reviewer #2 and shorten the evolutionary analysis which takes up more space than it adds to the value of the story.

We thank the reviewers and editor for the valuable suggestion. We re-structured the figures as suggested and rewrote the results section accordingly. The evolutionary analysis was significantly shortened.

1. Provide statistics for the imaging analysis such as TEM as only a single representative image is shown.

We agree that the observed morphological defects require a detailed statistical evaluation. TEM analysis was performed to confirm the results from optical microscopy and representative images with high magnification are shown for a detailed visualization of the defects. For additional quantification, we included statistics for IF stainings against calyx proteins CCIN and CapZa (Fig. 2 I-J). For TEM, we added additional images to the supplement (Figure 3 – supplement 1). Furthermore, we quantified the manchette length of step 10-13 spermatids to prove the increased elongation of the manchette in *Cylc2*^{-/-} and *Cylc1*^{-/y} *Cylc2*^{-/-} spermatids (Fig. 5 A-B).

1. Please consider other points raised by the reviewers below to improve the manuscript and provide responses on how the authors have addressed them.

We thank all reviewers for the detailed review of our manuscript and their valuable suggestions, which helped a lot to improve the manuscript. We considered all points raised by the reviewers to the best of our knowledge and hope that the reviewers will approve the manuscript ready for publication. We added a point-by-point discussion of all comments/suggestions hereafter.

Reviewer #1 (Recommendations For The Authors):

Major comments:

*(1) Antibody specificity: Fig 1E - there are some unspecific binding in *Cylc2*^{-/-} for CYLC2 and in *Cylc1*^{-/y} *Cylc2*^{+/-} for CYLC1 in the testis (and elongating spermatids in Figure 1 – Supplement 4). Could authors elaborate/comment on this? Western blot analysis would be also helpful to further support the antibody specificity.*

The very weak unspecific staining in the testis for CYLC2 (in *Cylc2*^{-/-}) and CYLC1 (in *Cylc1*^{-/y} *Cylc2*^{+/-}) is only present in the lumen of the seminiferous tubules and/or the residual bodies of the testicular sperm cells and can be referred to as background signal. Importantly, the signal is entirely lost in the PT region, proving specificity of the generated antibodies. We added the following paragraph to the results section:

Line 124-127: The generated antibodies did not stain testicular tissue and mature sperm of *Cylc1*- and *Cylc2*-deficient males, except for a very weak unspecific background staining in the lumen of seminiferous tubules and the residual bodies of testicular sperm (Fig. 1 F).

Specificity of antibodies was additionally proven by immunohistochemical staining, showing a specific staining only in testis sections but not in any other organ tested.

Line 115-117: Specificity of antibodies was proven by immunohistochemical stainings, showing a specific staining in testis sections only, but not in any other organ tested (Figure 1 – supplement 2)

To further verify the specificity of generated antibodies and the generation of functional knockout alleles, we additionally performed Western Blot analysis on cytoskeletal protein fractions, confirming the results of the IF staining. No unspecific bands were detected in the Western Blot, further supporting the notion that the weak unspecific signals in IF resemble staining artifacts.

The paragraph reads now as follows:

Line 127-132: Additionally, Western Blot analyses confirmed the absence of CYLC1 and CYLC2 in cytoskeletal protein fractions of the respective knockout (Fig. 1 G), thereby demonstrating i) specificity of the antibodies and ii) validating the gene knockout. Of note, the CYLC1 antibody detects a double band at 40-45 KDa. This is smaller than the predicted size of 74 KDa as, but both bands were absent in *Cylc1*^{-/-}. Similarly, the CYLC2 Antibody detected a double band at 38-40 KDa instead of 66 KDa. Again, both bands were absent in *Cylc2*^{-/-}.

(2) Please provide more interpretation of the gene dosage effect of Cylicin 2. It is not common to see a gene dosage effect in the sperm phenotype as transcripts and proteins can be shared between haploids due to syncytium formation during spermatogenesis.

We agree and we apologize for the misinterpretation. In *Cylc2*^{+/-} mice expression of *Cylc2* was reduced by half but there was no altered phenotype observed. The sentence now reads as follows:

Line 112: In *Cylc2*^{+/-} animals expression of *Cylc2* was reduced by 50 %.

(3) Line 194-196 - the authors say that the sperm are smaller, with shorter hooks and increased circularity of the nuclei, and reduced elongation. Are these statistically significant? There seems to be a high variation in the graph in S2D and the statistical analysis is not given.

We agree, performed statistical analyses, and highlighted significantly altered values for sperm head elongation and circularity in Figure 2 – Supplement 3.

(4) Line 153-164 It is interesting that the absence of Cylc2 affected many parts of sperm structure. I think some ratios of sperm always have a morphological defect in diverse ways, so it is hard to confirm the finding only with a single sperm image. I think that it will be important to do some statistical analysis or at the minimum show more TEM images from TEM.

We agree that the observed morphological defects require a detailed statistical evaluation. TEM analysis was performed to confirm the results from optical microscopy and representative images with high magnification are shown for a detailed visualization of the defects. For additional quantification, we included statistics for IF stainings against calyx

proteins CCIN and CapZa (Fig. 2 I-J). For TEM, we added additional images to the supplement (Figure 3 – Supplement 1).

(5) Line 236-242 - I believe that the phenotype described applies to the sperm from Cylc2^{-/-} and Cylc1^y Cylc2^{-/-} animals; however, I think that the Cylc1^{-y} Cylc2^{+/-} has a more subtle, intermediate phenotype between the WT and the genotypes missing both Cylc^{-/-} alleles.

We agree and we added a quantification of manchette length at step 10-13 to visualize the differences between the genotypes. The section reads now as follows: Line 268-272: Manchette length was measured starting from step 10 until step 13 spermatids and the mean was obtained, showing that the average manchette length was 76-80 nm in wildtype, Cylc1^{-Y} and Cylc2^{+/-} while for Cylc2^{-/-} and Cylc1^{-Y} Cylc2^{-/-} spermatids mean manchette length reached 100 nm (Fig. 5 B). Cylc1^{-Y} Cylc2^{+/-} spermatids displayed an intermediate phenotype with a mean manchette length of 86 nm.

(6) Since CYLC1 staining is absent in Fig 5B, does that mean that the mutation resulted in protein degradation/instability? Is RNA present? Additional biochemical studies of Cyclins demonstrating the deleterious nature of the mutations would strengthen the molecular pathogenesis of the human mutations.

Thank you for raising these important questions. The CYLC1 variant c.1720G>C is predicted to cause the amino acid substitution p.(Glu574Gln). It is, thus, conceivable that the RNA is present but either the protein is degraded or misfolded and, therefore, not detectable by IF. Unfortunately, for personal reasons of the patient, it is currently not possible to receive additional semen samples, preventing additional analyses of the semen, e.g. analysis of Cylicin transcript level.

(7) Strongly suggest shortening the evolutionary analysis - all the corresponding materials are in supplemental while texts are extensive- or even consider entirely omitting. It does not add a lot to the current study.

We agree that the evolutionary analysis was very detailed. However, we think that the results are important to understand the role of Cylicins for male reproduction in general. The results obtained from the mouse model might be transferable to other species, including humans. Further, the results present a possible explanation for the subfertility of Cylc1-deficient mice, in contrast to infertility of Cylc2-deficient males. We shortened the section, the paragraph reads as follows:

Line 287-302: To address why Cylc2 deficiency causes more severe phenotypic alterations than Cylc1 deficiency in mice, we performed evolutionary analysis of both genes. Analysis of the selective constraints on Cylc1 and Cylc2 across rodents and primates revealed that both genes' coding sequences are conserved in general, although conservation is weaker in Cylc1 trending towards a more relaxed constraint (Fig. 6). A model allowing for separate calculation of the evolutionary rate for primates and rodents, did not detect a significant difference between the clades, neither for Cylc1 nor for Cylc2, indicating that the sequences are equally conserved in both clades.

To analyze the selective pressure across the coding sequence in more detail, we calculated the evolutionary rates for each codon site across the whole tree. According to the analysis, 34% of codon sites were conserved, 51% under relaxed selective constraint, and 15% positively selected. For Cylc2, 47% of codon sites conserved, 44% under neutral/relaxed constraint, and 9% positively selected. Interestingly, codon sites encoding lysine residues, which are proposed to be of functional importance for Cylicins, are mostly conserved. For Cylc1, 17% of lysine residues are significantly conserved and 35% of significantly conserved codons encode

for lysine. For *Cylc2*, this pattern is even more pronounced with 27.9% of lysine codons being significantly conserved and 24.3% of all conserved sites encoding for lysine (Fig. 6).

Minor comments:

(1) Line 114, 115, 118 à Figure 1D is already well-described in the previous paragraph and thus redundant. Ref Fig 1D, E; but only figure E shows IF. Maybe supposed to be E and F or just 1E?

We apologize for the mix-up with the subfigures. The mentioned paragraph refers to Fig. 1 E-F, which was corrected accordingly.

Line 117-123: Immunofluorescence staining of wildtype testicular tissue showed presence of both, CYLC1 and CYLC2 from the round spermatid stage onward (Fig. 1 E). The signal was first detectable in the subacrosomal region as a cap-like structure, lining the developing acrosome (Fig. 1 E-F, Figure 1 – supplement 3). As the spermatids elongate, CYLC1 and CYLC2 move across the PT towards the caudal part of the cell (Figure 1 – supplement 4). At later steps of spermiogenesis, the localization in the subacrosomal part of the PT faded, while it intensified in the postacrosomal calyx region (Fig. 1 E-F).

(2) Figure 1F - Arguably, IF images show expression of both CYLC1 and CYLC2 to reach/include the acrosome/hook portion of the sperm head, but the diagram does not reflect that. Why is that?

We agree and apologize for the inconsistency. The illustration was adjusted according to the experimental data showing localization of Cylicins in the whole ventral part of the sperm.

(3) Line 124 - PAS staining mentioned on line 124, is not explained (Periodic acid Schiff staining) until line 605

We agree and introduced the abbreviation accordingly. The PAS staining was moved to Fig. 4. The paragraph reads now as follows:

Line 220-222: To study the origin of observed structural sperm defects, spermiogenesis of Cylicin deficient males was analyzed in detail. PNA lectin staining and Periodic Acid Schiff (PAS) staining of testicular tissue sections were performed to investigate acrosome biogenesis.

(4) Some figures are hard to read due to being very small (S1B, 3F).

We agree and we increased the figure size. For former Figure 3F (now figure 4A), insets with higher magnification of representative sperm were added. Insets are additionally shown in Figure 4 – Supplement 1 in higher resolution.

(5) Line 139 Please specify whether the sperm was capacitated or not.

Analysis of the flagellar beat was performed with non-capacitated sperm. We clarified this in the main text:

Line 203: The SpermQ software was used to analyze the flagellar beat of non-capacitated *Cylc2*^{-/-} sperm in detail 22.

As described in the Material and Methods section, sperm were only activated in TYH medium, prior to analysis:

Line 732-733: Sperm samples were diluted in TYH buffer shortly before insertion of the suspension into the observation chamber.

(6) Line 142-145; *The sentence is interrupted strangely, perhaps the authors meant to write: "Interestingly, we observed that the flagellar beat of Cylc2^{-/-} sperm cells was similar to wildtype cells, however, with interruptions during which midpiece and initial principal piece appeared stiff whereas high-frequency beating occurs at the flagellar tip"*

We corrected the sentence accordingly.

Line 206-208: Interestingly, we observed that the flagellar beat of Cylc2^{-/-} sperm cells was similar to wildtype cells, however, with interruptions during which midpiece and initial principal piece appeared stiff whereas high frequency beating occurs at the flagellar tip (Fig. 3 C, Video 1, Video 2).

(7) Line 142 -Wrong Figure number. Figure S4A is a phylogenic analysis.

We regret the mix up and corrected the Figure reference accordingly. Line 204-205: Cylc2^{-/-} sperm showed stiffness in the neck and a reduced amplitude of the initial flagellar beat, as well as reduced average curvature of the flagellum during a single beat (Figure 3 – supplement 2).

(8) L146-147 Better placed in Discussion.

We agree, and we omitted this sentence from the results part.

(9) Line 154-156 - *The white arrowheads are present in both WT and KO sperm, thus it appears they denote the basal plate, not necessarily the dislocation/parallel position as the current text seems to suggest. Furthermore, the position of the WT and KO sperm is somewhat different with the tail coiling differently, so it is hard to see whether the two are comparable.*

We agree and we removed the white arrowhead in the WT sperm picture, and it now depicts only the dislocation of the basal plate in the Cylc2^{-/-} sperm. Due to the morphological anomalies of Cylc2^{-/-} sperm cells, it's difficult to determine the exact angle of the depicted cell. However, we added more TEM pictures of the sperm cells (3 for WT and 6 for Cylc2^{-/-}) in Figure 3 – Supplement 1.

(10) Line 164 Please describe in detail what mitochondrial damage the readers expect to see from the TEM image.

We evaluated the observed mitochondrial damage in more detail. Unfortunately, the defects described initially seem to be an artifact of apoptotic sperm cells and could not be identified in vital sperm cells in either of the knockout mouse models. We apologize for this misinterpretation, and we deleted this section in the manuscript.

(12) Figure S2A - no WT comparison, difficult to compare without it (mitochondria in Cylc2^{-/-})

See (10). We evaluated the observed mitochondrial damage in more detail and in comparison to WT. Unfortunately, the defects described initially seem to be an artifact of apoptotic sperm cells and could not be identified in vital sperm cells in either of the knockout mouse models. We apologize for this misinterpretation and we deleted this section in the manuscript.

(13) Line 172-173 - Fig 3C denotes quantification of abnormal acrosome only, however, the text mentions sperm coiled tail being quantified within this graph - which is it? Is it both of them? Or only one of them?

Figure 3 C (now Figure 2G) showed the percentage of abnormal sperm in general comprising acrosomal as well as flagellar defects. We modified the figure and evaluated acrosomal defects and tail defects separately. The results section was changed accordingly and reads now as follows:

Line 152-159: Loss of *Cylc1* alone caused malformations of the acrosome in around 38% of mature sperm, while their flagellum appeared unaltered and properly connected to the head. *Cylc2*^{+/−} males showed normal sperm tail morphology with around 30% of acrosome malformations. *Cylc2*^{−/−} mature sperm cells displayed morphological alterations of head and mid-piece (Fig. 2 F-G). 76% of *Cylc2*^{−/−} sperm cells showed acrosome malformations, bending of the neck region, and/or coiling of the flagellum, occasionally resulting in its wrapping around the sperm head in 80% of sperm (Fig. 2 F). While 70% of *Cylc1*^{−/−} *Cylc2*^{+/−} sperm showed these morphological alterations, around 92% of *Cylc1*^{−/−} *Cylc2*^{−/−} sperm presented with coiled tail and abnormal acrosome (Fig. 2 F-G).

(14) Fig 3D - CCIN in the text, cylicin in the figure - this should be consistent. Furthermore, since only the head is being shown, is CCIN ever detected in the WT sperm tail?

We apologize for the inconsistency, and we added the abbreviation “CCIN” to the figure. CCIN is solely detectable in the sperm head of wildtype sperm as published previously. Irregular staining patterns showing signals in the tail region are only observed upon Cylicin deficiency.

(15) Line 199-200 - To say that head of Cylc2-deficient sperm appears less concave seems redundant, likely the observed increased circularity is contributed to by sperm head being less concave in this region; unless there is an extra point that the authors are trying to make and if so, this needs to be elaborated on

We agree and we deleted the sentence from the manuscript.

(16) Figure legend of Fig S3 is wrong. Only S3A and S3B are present, and in the figure legend S3C corresponds to figure S3B.

We agree and corrected the Figure legends accordingly. Due to the re-structuring of the manuscript, Figures and Supplementary figures were re-ordered as well.

(17) Figure 4B - figure legend and/or text should specify that lectin is green and HOOK1 is in red

We specified the figure legend as well as the main text accordingly: Line: 279-281: Co-staining of the spermatids with antibodies against PNA lectin (green) and HOOK1 (red) revealed that abnormal manchette elongation and acrosome anomalies simultaneously occurred in elongating spermatids of *Cylc2*^{−/−} male mice (Fig. 5 C).

Line: 560-562: Co-staining of the manchette with HOOK1 (red) and acrosome with PNA-lectin (green) is shown in round, elongating and elongated spermatids of WT (upper panel) and *Cylc2*^{−/−} mice (lower panel).

(18) Line 261-263 - It is difficult to see what is going on with microtubules in these images, as the resolution is low

We increased the pictures and improved their quality. Microtubules are also depicted with letter ‘m’

(19) Line 265-266 - It seems that there is a persistence of manchette, rather than elongation. From these images, I cannot see gaps, and I am not sure where to look for them. This needs to be labelled further and higher-resolution images could be included for clarity.

We agree, although we observed both excessive elongation and persistence of the manchette. The average length of the manchette is now shown in figure 5B.

The paragraph now reads as follows:

Line 235-239: Microtubules appeared longer on one side of the nucleus than on the other, displacing the acrosome to the side and creating a gap in the PT (Fig. 4 C). Whereas elongated spermatids at step 14-15 in wildtype sperm already disassembled their manchette and the PT appeared as a unique structure that compactly surrounds nucleus, in *Cylc2*^{-/-} spermatids, remaining microtubules failed to disassemble.

The gaps in the perinuclear theca are better visible in TEM micrographs and the description is now in the paragraph describing TEM.

(20) Line 269 Please include the information of "White arrowhead".

We added the information accordingly.

Line 240-242: In addition, at step 16, the calyx was absent, and an excess of cytoplasm surrounded the nucleus and flagellum (Fig. 4 C, white arrowhead).

(21) Line 276-280 This should be placed in the Discussion.

We agree, and we deleted this concluding remark from the results section.

(22) Is Cylc1 and/or Cylc2 conserved/expressed amongst species other than rodents and primates?

Yes, *Cylc1* and *Cylc2* homologs were identified in *C. elegans* for example. We added a schematic to the introduction showing the protein structure of human, mouse and *C. elegans* CYLC1 and CYLC2 (Figure 1 – supplement 1).

The section reads now as follows:

Line 73-78: In most species, two Cylicin genes, *Cylc1* and *Cylc2*, have been identified (Figure 1-supplement 1). They are characterized by repetitive lysine-lysine-aspartic acid (KKD) and lysine-lysine-glutamic acid (KKE) peptide motifs, resulting in an isoelectric point (IEP) > pH 10 14, 15. Repeating units of up to 41 amino acids in the central part of the molecules stand out by a predicted tendency to form individual short α -helices 14. Mammalian Cylicins exhibit similar protein and domain characteristics, but CYLC2 has a much shorter amino-terminal portion than CYLC1 (Figure 1-supplement 1).

(23) The whole chapter "Cylc2 coding sequence is slightly more conserved among species than Cylc1" references only supplemental figures/tables. I find this unusual.

We agree, and in order to show the results of the evolutionary analysis more clearly, we moved the panel to main Figure 6.

Line 286-302: To address why Cylc2 deficiency causes more severe phenotypic alterations than Cylc1 deficiency in mice, we performed evolutionary analysis of both genes. Analysis of the selective constraints on Cylc1 and Cylc2 across rodents and primates revealed that both genes' coding sequences are conserved in general, although conservation is weaker in Cylc1 trending towards a more relaxed constraint (Fig. 6 A). A model allowing for separate calculation of the evolutionary rate for primates and rodents, did not detect a significant difference between the clades, neither for Cylc1 nor for Cylc2, indicating that the sequences are equally conserved in both clades.

To analyze the selective pressure across the coding sequence in more detail, we calculated the evolutionary rates for each codon site across the whole tree. According to the analysis, 34% of codon sites were conserved, 51% under relaxed selective constraint, and 15% positively selected. For Cylc2, 47% of codon sites conserved, 44% under neutral/relaxed constraint, and 9% positively selected. Interestingly, codon sites encoding lysine residues, which are proposed to be of functional importance for Cylicins, are mostly conserved. For Cylc1, 17% of lysine residues are significantly conserved and 35% of significantly conserved codons encode for lysine. For Cylc2, this pattern is even more pronounced with 27.9% of lysine codons being significantly conserved and 24.3% of all conserved sites encoding for lysine (Fig. 6 B).

(24) Line 332 - CYCL2 should be CYLC2

We corrected the typo accordingly.

(25) Line 340 The ratio of head defects is different from Table 1 (98% here and 99 % in the table). Please check this information.

We apologize for the inconsistency. We checked the raw data and corrected the table accordingly.

(26) Line 344-345 From figure 5C it is difficult to determine whether the sperm are "headless" or whether the heads are attached to the highly coiled tails next to them

We agree and we quantified the percentage of sperm showing abnormal flagella and a headless phenotype. Furthermore, we added an arrowhead to figure 6C to highlight headless sperm. The paragraph reads now as follows:

Line 335-339: Bright field microscopy demonstrated that M2270's sperm flagella coiled in a similar manner compared to flagella of sperm from Cylicin deficient mice. Quantification revealed 57% of M2270 sperm displaying abnormal flagella compared to 6% in the healthy donor (Fig. 7 D). Interestingly, DAPI staining revealed that 27% of M2270 flagella carry cytoplasmic bodies without nuclei and could be defined as headless spermatozoa (Fig. 7 C, white arrowhead; Fig. 7 E).

(27) L367-368 I agree with the authors' logic of this sentence. Although, it is better to show the co-localization of proteins using multi-channel immunocytochemistry. As you mentioned on L369 this will make your finding more obvious. If it is available, please include the colocalization images of the proteins.

We performed the multi-channel staining against Cylicin1 and Calicin, as well as Cylicin2 and Calicin on mouse epididymal sperm and it's shown in Figure 2 – supplement 4. Unfortunately, we did not manage to obtain stainings of tissue sections since antibodies against Cylicins and Calicin require different sample processing.

The sentence was added in the section describing calyx integrity:

Line 168-169: In epididymal sperm, CCIN co-localizes with both CYLC1 and CYLC2 in the calyx (Figure 2 – supplement 4).

(28) Line 376 Please keep the abbreviation. "Calicin" "CCIN".

We included the abbreviation accordingly.

Line 377-378: CCIN is shown to be necessary for the IAM-PT-NE complex by establishing bidirectional connections with other PT proteins.

(29) Line 377-378 "Based on ~". The authors did not prove the interaction between CCIN and Cylicins in this article. The mislocalization of CCIN might be resulted in the loss of Cylicins, without any "interaction". To reach this conclusion, a more direct result should be provided.

We agree that we overinterpreted this as we and others did not prove the interaction between CCIN and Cylicins so far. We therefore weakened this statement and formulated it as a hypothesis.

Line 377-381: CCIN is shown to be necessary for the IAM-PT-NE complex by establishing bidirectional connections with other PT proteins. Zhang et al. found CYLC1 to be among proteins enriched in PT fraction 7. Based on their speculation that CCIN is the main organizer of the PT, we hypothesize that both CCIN and Cylicins might interact, either directly or in a complex with other proteins, in order to provide the 'molecular glue' necessary for the acrosome anchoring.

(30) Line 499 Please specify which is the target of the immunostaining on the Figure legend. (Tubulin à acetylated α -tubulin)

We specified that α -Tubulin was stained. The figure legend reads now as follow: Line 555-557: Immunofluorescence staining of α -Tubulin to visualize manchette structure in squash testis samples of WT, Cylc1-/-, Cylc2+/-, Cylc2-/-, Cylc1 -/y Cylc2+/- and Cylc1-/- Cylc2-/- mice.

(31) Line 502 Please specify which color indicates which target protein (not only cellular structure).

Line 560-562: Co-staining of the manchette with HOOK1 (red) and acrosome with PNA-lectin (green) is shown in round, elongating and elongated spermatids of WT (upper panel) and Cylc2-/- mice (lower panel).

(32) Line 509 Please include scale bar information in the figure legend like Figure 4 (The magnifications of Figure 5 B, C, and D seem different).

We included the scale bar information accordingly (now Figure 6).

Line 575-588: Figure 6: Cylicins are required for human male fertility

(A) Pedigree of patient M2270. His father (M2270_F) is carrier of the heterozygous CYLC2 variant c.551G>A and his mother (M2270_M) carries the X-linked CYLC1 variant c.1720G>C in a heterozygous state. Asterisks (*) indicate the location of the variants in CYLC1 and CYLC2 within the electropherograms.

(B) Immunofluorescence staining of CYLC1 in spermatozoa from healthy donor and patient M2270. In donor's sperm cells CYLC1 localizes in the calyx, while patient's sperm cells are completely missing the signal. Scale bar: 5 μ m.

(C) Bright field images of the spermatozoa from healthy donor and M2270 show visible head and tail anomalies, coiling of the flagellum as well as headless spermatozoa who carry cytoplasmic residues without nuclei. Heads were counterstained with DAPI. Scale bar: 5 μ m.

(D-E) Quantification of flagellum integrity (D) and headless sperm (E) in the semen of patient M2270 and a healthy donor.

(F-G) Immunofluorescence staining of CCIN (F) and PLCz (G) in sperm cells of patient M2270 and a healthy donor. Nuclei were counterstained with DAPI. Scale bar: 3 μ m.

(33) S2A is not clear. Please describe specifically what the left panel and right panel are about to show with a clear indication of what is PM, mitochondria, etc. On the right, in only one cross-section that shows both mitochondria and the 9+2 axoneme, they look both unaltered whereas on the left, there are unpacked, not aligned mitochondria but the tail boundary is not clear to grasp at first sight.

We apologize for the bad quality of the TEM pictures showing the axonemes and the missing labeling. We recorded and included new images showing an intact 9+2 microtubular structure in Cylc2^{-/-}. Furthermore, we added an image for the wildtype control.

(34) S2D: colors of the last three plots of each graph are too close to tell apart

We agree and changed the color scheme for better visualization.

Reviewer #2 (Recommendations For The Authors):

However, I find the manuscript a bit messy, and I will propose to reorganize the figures: following figure 1, showing the reproductive phenotype, I would continue with a figure showing the morphology of sperm in optical microscopy and showing the morphological defect of the nucleus (Fig 3B and 3E), followed with one figure focusing on the flagellum, with images obtained with optical and electronic microscopies, allowing to present the abnormalities of the flagellum and finally the impact on sperm motility and flagellum beating (mix of figure 2FG/3A); next, one figure focusing on acrosome. After that, I would present all data concerning spermiogenesis, starting with figure 2C.

We thank the reviewer for the valuable suggestion, which helps a lot to improve the structure and comprehensibility of the manuscript. We re-organized the figures and the results section accordingly.

Major remarks

1. Line 111. The specificity of raised Ab is not clear. Please specify if antibodies are specific: what immune-decorates anti-CYLC1: only CYLC1 or CYLC1 and CYLC2. Same question for anti-CYLC2

Both antibodies were raised against specific peptides of the CYLC1 or CYLC2 protein, respectively. The antigen peptides used for immunization are depicted in the Material and Methods section (AESRKSNDERRKTLKIKFRGK and KDAKKEGKKKGKRESRKKR peptides for CYLC1; KSVGTHKSLASEKTKKEVK and ESGGEKAGSKKEAKDDKKDA for CYLC2). The peptides used for immunization are specific as they do not resemble the highly conserved and repetitive KKD/KKE motives present in both, Cylc1 and Cylc2.

The specificity of raised antibodies was validated by IF staining of wildtype and Cylicin-deficient testis sections. The results clearly show, that CYLC1 signal is absent in Cylc1-deficient spermatids and CYLC2 signal being absent in Cylc2 deficient spermatids.

Specificity of antibodies was additionally proven by immunohistochemical stainings, showing a specific staining only in testis sections but not in any other organ tested.

Line 115-117: Specificity of antibodies was proven by immunohistochemical stainings, showing a specific staining only in testis sections but not in any other organ tested (Figure 1 - supplement 2)

To further verify the specificity of generated antibodies and the generation of functional knockout alleles, we additionally performed Western Blot analysis on cytoskeletal protein fractions, confirming the results of the IF staining.

The paragraph reads now as follows:

Line 127-134: Additionally, Western Blot analyses confirmed the absence of CYLC1 and CYLC2 in cytoskeletal protein fractions of the respective knockout (Fig. 1 G), thereby demonstrating i) specificity of the antibodies and ii) validating the gene knockout. Of note, the CYLC1 antibody detects a double band at 40-45 KDa. This is smaller than the predicted size of 74 KDa as, but both bands were absent in Cylc1-/-y. Similarly, the CYLC2 Antibody detected a double band at 38-40 KDa instead of 66 KDa. Again, both bands were absent in Cylc2-/-y. Next, Mass spectrometry analysis of cytoskeletal protein fraction of mature spermatozoa was performed detecting both proteins in WT but not in the respective knockout samples (Figure 1 – supplement 5; Figure 1 – supplement 6).

1. Line 115 and figure 1D. From the images presented in figure 1D, it is not clear where CYLC1 and CYLC2 are localized in the round and in elongated spermatids. Please make double staining using a second Ab to identify the acrosome such as DPY19L2 (best option) or SP56 and the manchette such as acetylated alpha-tubulin.

We agree, and we added a double staining of CYLC1/CYLC2 and SP56 to the supplement (Figure 1 - supplement 3), showing co-localization of the developing acrosome and Cylicins. Manchette staining was not performed due to antibodies being available in same species as those against Cylicins (anti-rabbit).

Line 117-120: Immunofluorescence staining of wildtype testicular tissue showed presence of both, CYLC1 and CYLC2 from the round spermatid stage onward (Fig. 1 E, Figure 1 – supplement 3). The signal was first detectable in the subacrosomal region as a cap like structure, lining the developing acrosome (Fig. 1 E-F, Figure 1 – supplement 3).

1. Line 118 and figure 1. The drawing showing the localization of Cylicin in mature sperm does not fit with the experimental data. Cylicins are located on the whole ventral face of the sperm.

We agree and apologize for the inconsistency. The illustration was adjusted according to the experimental data showing localization of Cylicins in the whole ventral part of the sperm.

1. Figure 1: Change "expression of Cylicin" to "localization of cylicin" (green)

We changed the legend accordingly.

1. Line 124 and figure 2C. In the figure provided, the PAS staining seems defective. The acrosomes do not seem stained (in pink as expected for a PAS staining). It may be due to the low quality of the pdf file, nevertheless, it is important to provide in supplementary data, an enlargement of the spermatid region showing the staining of the acrosome.

We apologize for the bad quality of the PDF file and the low magnification. We restructured the subfigure showing PAS stained spermatids at different steps of spermiogenesis at higher magnification. According to the initial reviewer's suggestion, the PAS staining was moved to figure 4. The PAS staining in figure 2 was replaced by HE-stained overview testis sections in Figure 3 – supplement 1 showing intact spermatogenesis in all genotypes.

1. Line 130. Please indicate a reference for the lower limit of 58%. If this lower limit corresponds to human sperm, it should be omitted.

Indeed, the given reference limit of 58% is only valid for human sperm samples. Therefore, we omitted the reference limit. The paragraph reads now as follows: Line 144-146: Eosin-Nigrosin staining revealed that the viability of epididymal sperm from all genotypes was not severely affected (Fig. 2 D, Figure 2 – supplement 2).

1. line 152 Sperm morphology. Before showing the ultrastructure of the sperm, it would be important to show sperm morphology observed by optical microscopy. Therefore, I recommend including figure S2 as a principal figure, with a mix of Figures 3B and 3E.

We thank the reviewer for the suggestion. The results section was re-structured accordingly, first showing results of optical microscopy (Fig. 2), followed by an in-depth ultrastructural investigation of morphological defects and their effects on sperm motility. Brightfield images of epididymal sperm were moved from former Figure S2 to main Figure 2.

1. Line 164. figure S2A, showing that the 9+2 pattern is normal in KO sperm, is not convincing. Enlargement is required to conclude that the axoneme structure is normal; from the pictures, it rather seems that some doublets are missing.

We apologize for the bad quality of the TEM pictures showing the axonemes. We recorded and included new images showing an intact 9+2 microtubular structure.

1. Line 196. I would suggest removing the term "mild globozoospermia". Globozoospermia is rather complete (100% of round sperm heads) or incomplete (<90 % of round sperm heads). The anomalies observed on sperm heads, sperm motility, and the decrease in sperm concentration are rather suggestive of an OAT.

We agree and we omitted the term "mild globozoospermia". Instead, we added a concluding remark to the section, summarizing the described defects as OAT syndrome. The section reads now as follows:

Line 215-217: Taken together, observed anomalies of sperm heads, impaired sperm motility, and the decrease in epididymal sperm concentration show that Cylc deficiency results in a severe OAT phenotype (Oligo-Asthenozoospermia-syndrome) described in human.

1. Line 248. It is not clear from the data of figure 4B that "the developing acrosome lost its compact adherence to the nuclear envelope". From this figure, only defective morphologies of the acrosome are observed

We agree and we omitted the sentence. Furthermore, it does not add additional information to the manuscript, since defects in the attachment of the acrosome to the nuclear envelope have been described in detail in Figure 4C.

1. line 260-264. *Manchette defects appear at stages 9-10. At this stage, the HTCA is already attached to the nucleus of the spermatid. see for instance figure 2 from Shang Y, Zhu F, Wang L, Ouyang YC, Dong MZ, Liu C, Zhao H, Cui X, Ma D, Zhang Z, Yang X, Guo Y, Liu F, Yuan L, Gao F, Guo X, Sun QY, Cao Y, Li W. Essential role for SUN5 in anchoring sperm head to the tail. Elife. 2017 Sep 25;6:e28199. doi: 10.7554/eLife.28199 . Therefore, the hypothesis that "abnormal attachment of the developing flagellum to the basal plate and consequently flipping of the head and coiling of the tail in mature spermatozoa" is unlikely and I suggest modifying this paragraph. In the HOOK paper, the manchette defects occurred earlier.*

We read the suggested literature and we agree to this reviewer's comment. Manchette defects that we observe appear at later stages and are probably not responsible for the morphological anomalies of the mature sperm. We also re-analyzed all the manchette staining pictures and didn't find any defects at earlier stages, so we decided to delete the sentence from the manuscript.

1. Line 344. *Please indicate a percentage of headless spermatozoa. Many sperm is too vague.*

We agree and we quantified the percentage of sperm showing abnormal flagella and a headless phenotype. The paragraph reads now as follows:

Line 335-339: Bright field microscopy demonstrated that M2270's sperm flagella coiled in a similar manner compared to flagella of sperm from Cylicin deficient mice. Quantification revealed 57% of M2270 sperm displaying abnormal flagella compared to 6% in the healthy donor (Fig. 7 D). Interestingly, DAPI staining revealed that 27% of M2270 flagella carry cytoplasmatic bodies without nuclei and could be defined as headless spermatozoa (Fig. 7 C, white arrowhead; Fig. 7 E).

1. *Any data concerning the success of ICSI for this patient?*

Yes, the outcome of the ICSI were added to the main text. Line 309-311: The couple underwent one ICSI procedure which resulted in 17 fertilized oocytes out of 18 retrieved. Three cryo-single embryo transfers were performed in spontaneous cycles, but no pregnancy was achieved.

1. *Finally, it would be interesting to study the localization of PLCzeta in this model, since its localization in the perinuclear theca has been clearly shown (Escoffier et al, 2015 doi:10.1093/molehr/gau098)*

We thank the reviewer for the valuable suggestion and performed PLCzeta staining on human sperm, clearly showing an irregular PT staining pattern in sperm of patient M2270 compared to healthy control sperm. Of note, staining was not possible in the mouse due to the antibody being reactive only for human samples.

The section reads as follows:

Line 343-349: Testis specific phospholipase C zeta 1 (PLCζ1) is localized in the postacrosomal region of PT in mammalian sperm (Yoon and Fissore, 2007) and has a role in generating

calcium (Ca^{2+}) oscillations that are necessary for oocyte activation (Yoon, 2008). Staining of healthy donor's spermatozoa showed a previously described localization of PLC ζ 1 in the calyx, while sperm from M2270 patient presents signal irregularly through the PT surrounding sperm heads (Fig. 7 G). These results suggest that Cylicin deficiency can cause severe disruption of PT in human sperm as well, causing male infertility.

Reviewer #3 (Recommendations For The Authors):

1. *Why the Cylc1-/-y Cylc2+/- males were infertile? It would be helpful to show the homologue of the two proteins;*

To elaborate more on the homology of CYLC1 and CYLC2, we added a more detailed section about the protein and domain structure to the introduction.

Line 73-78: In most species, two Cylicin genes, Cylc1 and Cylc2, have been identified (Figure 1supplement 1). They are characterized by repetitive lysine-lysine-aspartic acid (KKD) and lysine-lysine-glutamic acid (KKE) peptide motifs, resulting in an isoelectric point (IEP) > pH 10 14, 15. Repeating units of up to 41 amino acids in the central part of the molecules stand out by a predicted tendency to form individual short α -helices (Hess et al., 1993). Mammalian Cylicins exhibit similar protein and domain characteristics, but CYLC2 has a much shorter amino-terminal portion than CYLC1 (Figure 1supplement 1).

Speculations about the infertility of Cylc1-/-y Cylc2+/- males was added to the discussion:

Line 410-413: Interestingly, Cylc1-/-y Cylc2+/- males displayed an “intermediate” phenotype, showing slightly less damaged sperm than Cylc2-/- and Cylc1-/-y Cylc2-/- animals. This further supports our notion, that loss of the less conserved Cylc1 gene might be at least partially compensated by the remaining Cylc2 allele.

1. *Western blot is important to show the absence of the two proteins in the mouse models;*

To further verify the specificity of generated antibodies and the generation of functional knockout alleles, we additionally performed Western Blot analysis on cytoskeletal protein fractions, confirming the results of the IF staining.

A paragraph was added to the manuscript and reads as follows:

Line 127-134: Additionally, Western Blot analyses confirmed the absence of CYLC1 and CYLC2 in cytoskeletal protein fractions of the respective knockout (Fig. 1 G), thereby demonstrating i) specificity of the antibodies and ii) validating the gene knockout. Of note, the CYLC1 antibody detects a double band at 40-45 KDa. This is smaller than the predicted size of 74 KDa as, but both bands were absent in Cylc1-/-y. Similarly, the CYLC2 Antibody detected a double band at 38-40 KDa instead of 66 KDa. Again, both bands were absent in Cylc2-/-y. Next, Mass spectrometry analysis of cytoskeletal protein fraction of mature spermatozoa was performed detecting both proteins in WT but not in the respective knockout samples (Figure 1 – supplement 5; Figure 1 – supplement 6).

1. *On Page 7, line 227 and line 243, was the acetylated α -tubulin or α -tubulin antibody used?*

For all stainings α -tubulin antibody was used. We corrected this accordingly. Line 257-259: We used immunofluorescence staining of α -tubulin on squash testis samples containing spermatids at different stages of spermiogenesis to investigate whether the altered head

shape, calyx structure, and tail-head connection anomalies originate from possible defects of the manchette structure.

1. *Fig. 2S: A cartoon showing the elongation and circularity of nuclei for evaluation is helpful; The TEM images from the control and Cylc1 KO mice are needed;*

Cylc1-/Y TEM picture was added in Figure 3A.

1. *The discussion should be rewritten. The current version is to repeat the experiments/findings. The authors should discuss more about the potential mechanisms.*

We discussed the observed defects of Cylc-deficient animals and discussed this in relation to other published mouse models deficient in Calyx components. Furthermore, we speculated about potential interaction partners of Cylicins and the importance of these protein complexes for male fertility. However, to this point, we think that it is too farfetched to speculate about potential mechanisms without any evidence for Cylc interaction partner or their exact molecular function. This requires further research.